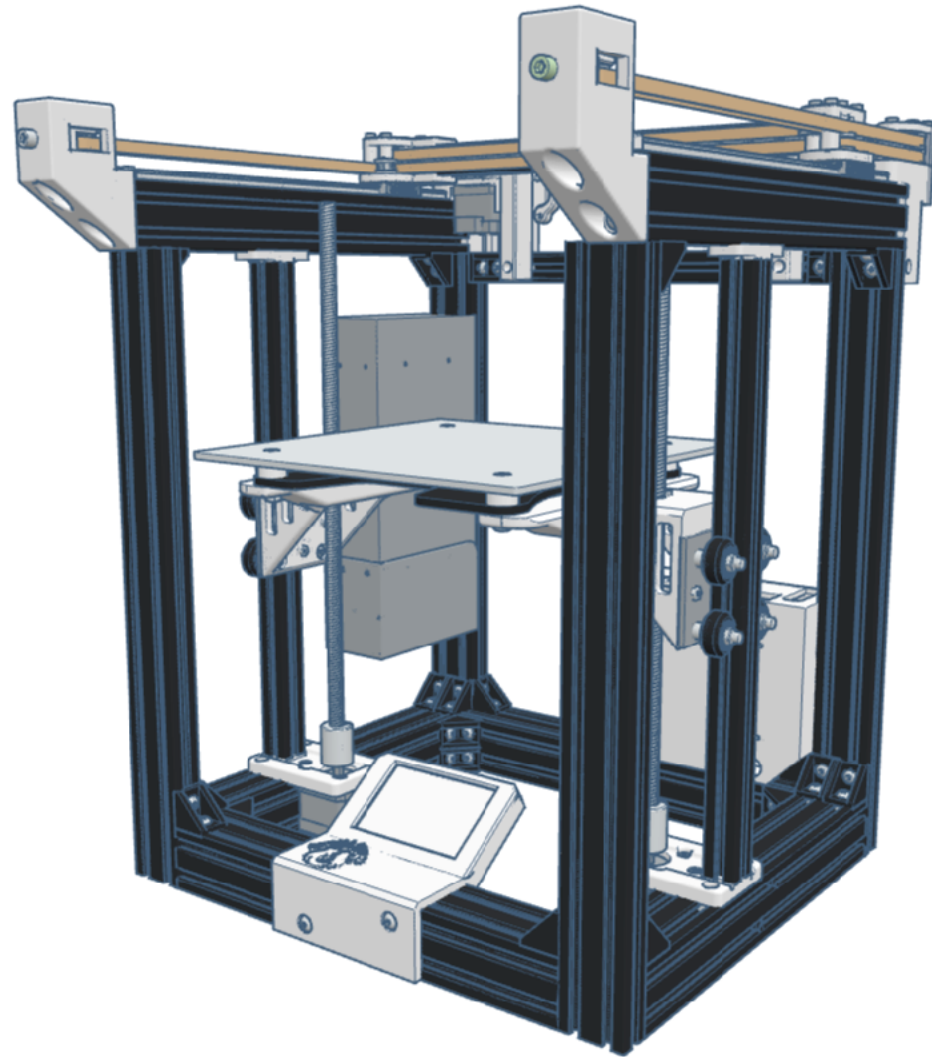




**Duender - MGN9H
V-wheels Z axis**



Your Open-Source 2 x Creality Ender-3 core XY conversion

by Sergei Irbis

Some points of the build require your attention



Once the printer is complete, the print bed and the nozzle will be heating and may cause **severe burning** if you are not cautious.

Do not touch those parts while in function.



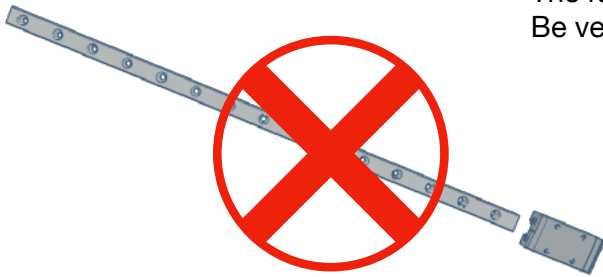
A risk of overheating or electric short in components is always possible. In some cases, this could lead to the printer **starting a fire**.

Please, double check your build and babysit your printer during the tuning phase as well as its firsts jobs

Keeping an **extinguisher** near those appliances is a good practice.

Read the full doc before starting your build, and do not hesitate to reach out on the [Discord server](#) to ask for help any time you are unsure

Linear rails handling



The rail axis carriages are meant to slide easily.

Be very careful **they do not fall off the rail**, as this would most likely result in irreparable damage.

The MGN9H rails are quite thin and can be somewhat difficult to fit onto V-slot extrusions.

You can use some **3D printed helpers** to align them: use [spacers](#) or tools like [this](#) or [this](#).

Try tightening the screws progressively along the rail while maintaining alignment until all screws are secured.

When you receive them, they are very often covered in some kind of protective oil.

The **video below by Canuck Creator** offers all the advice you need to clean and lubricate your rails.



<https://2cm.es/1ml5J>

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Need more information?



<https://2cm.es/1h2CJ>



<https://2cm.es/1h2CE>



<https://2cm.es/1IYzA>

Wanna support the project?

If you enjoy Duender and want to support what I do, there are a few ways to help:

Patreon — monthly support

<https://patreon.com/Irbis3D>

Ko-fi / Buy Me a Coffee — one-time donations

<https://buymeacoffee.com/Irbis3D>

YouTube Super Thanks — under any video

<https://youtube.com/@Irbis3D>

Your support helps me create more content, upgrade gear, and keep experimenting with cool ideas.

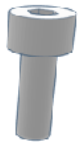
Thank you.

Sergei Irbis.

Know your bits

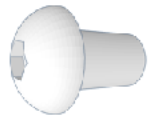
Hardware reference

1



SHCS (Socket Head Cap Screw)

Metric fastener with cylindrical head and hex drive



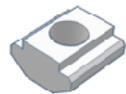
BHCS (Button Head Cap Screw)

Metric fastener with hemispheric head and hex drive. Mostly used with M5 T-slot nuts



FHCS (Flat Head Cap Screw)

Metric fastener with cone shaped head and flat top



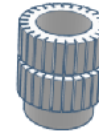
T-slot nut

Nut that is inserted inside the slot of aluminium extrusions. Must be inserted on beforehand
Hammerhead T-nuts can be used after assembly



Hex nut

Coupled with a screw to create a bolt. Nylstop ones have a plastic insert to prevent them un-tightening



Heat-set insert

Once heated with the soldering iron, it melts into plastic, which makes a tight fit by solidifying.
M3 threading, 5 mm outer diam., 4 mm



Shim (or spacer)

Metal flat ring, with a set thickness, used as spacer between parts and bearings. Not to be confused with washers



Pin

Metal smooth rod but to size. Used as an axis for idlers.



Toothed idler

Has bearings inserted in the axis, used to redirect the belt in the motion system. GT2 refers to the belt pitch between teeth



Toothless idler

Similar idler without teeth, for the smooth side of the belt. 16T is for teeth count, 6W for belt width, 3B for axis bore diameter

BOM and sourcing

Bill of materials

2

Fasteners*	Qty	Order	Affiliated link / Notes	Hardware	Qty	Order	Affiliated link
M5x25 SHCS	2	✓	From Ender 3	42-34 Nema17 motor	2	✓	From Ender 3
M4x20 SHCS	2	✓	From Ender 3	42-40 Nema17 motor	2	✓	From Ender 3
M5x30 BHCS	8	✓	From Ender 3	Leadscrew T8x8 mm 365 mm	2	✓	From Ender 3
M4x35 FHCS	4	✓	From Ender 3	Eccentric column D7.2 d5 H8.35 mm	4	✓	From Ender 3
M5 hex locknut	10	✓	From Ender 3	Spacer for rollers D8 d5 H8.35 mm	4	✓	From Ender 3
Corner bracket	28		https://s.click.aliexpress.com/e/_c4DUW5sX	V-wheels rollers	8	✓	From Ender 3
M3x30 SHCS	22			Y plate	1	✓	From Ender 3
M3x20 SHCS	2		https://s.click.aliexpress.com/e/_c2Rkq5Jx	Motor coupler D5x8 <i>optional elastic coupler</i>	2 2	✓	From Ender 3
M3x18 SHCS	8			Leadscrew brass nut <i>Optional anti backlash nut</i>	2 2	✓	From Ender 3
M3x16 SHCS	6			2020 V-slot 300mm extrusion	1		If you install the front brace (see p.15)
M3x8 SHCS	32			MGN9H 300mm rail	3		https://s.click.aliexpress.com/e/_oFYW0St
M3x6 SHCS	8			GT2 6mm wide belt	5m		https://s.click.aliexpress.com/e/_ooDLTEH
M5x8 BHCS	94			Pulley GT2 16T 6W 5B	2		https://s.click.aliexpress.com/e/_c4mjCdxl
M5 T-slot nut	79		https://s.click.aliexpress.com/e/_c4EDoXzD	Toothed idler GT2 16T 6W 3B	2		https://s.click.aliexpress.com/e/_c3Fqpe71
M3 T-slot nut	26		https://s.click.aliexpress.com/e/_c4EDoXzD	Toothless Idler GT2 16T 6W 3B	10		https://s.click.aliexpress.com/e/_c3Fqpe71
M4 hex locknut	4			M3x0,5 mm shim	48		https://s.click.aliexpress.com/e/_c4BNsuvz
M3 hex nut	20			M3x10 mm spacer	1		https://s.click.aliexpress.com/e/_c3rUz3wF
M3 heat-set insert M3x5x4	28		https://s.click.aliexpress.com/e/_c4DbNkaX				

Long screws kit: https://s.click.aliexpress.com/e/_c4VYXcQR (colour: Button head)

Short screws kit: https://s.click.aliexpress.com/e/_c30o44Xt

* Note that most of the time, SHCS can be replaced by BHCS and vice-versa. Please check the manual to see where it may apply.

BOM and sourcing

3

Electronics	Qty	Order	Affiliated link
24V 15A PSU w/ housing, 10A switch, XT60 cable	1	✓	From Ender 3
Heatbed 24V 235x235 mm	1	✓	From Ender 3
4010 24V axial fan	2	✓	From Ender 3
12864 screen w/ support plate <i>Optional BTT HDMI5 V1.2</i>	1 1	✓	From Ender 3 https://s.click.aliexpress.com/e/_c4odBUeJ
BTT MANTA 5P	1		https://s.click.aliexpress.com/e/_oktZaKt
BTT CB2	1		https://s.click.aliexpress.com/e/_oDMokmp
BTT TMC 2209	5		https://s.click.aliexpress.com/e/_oktZaKt
Misc	Qty	Order	Affiliated link
12 mm silicone bed spacer <i>Optional Spring bed spacer</i>	4 4	✓	From Ender 3
Aluminium 60x66 mm 3 mm thick <i>Optional Open build Plates</i>	2 2		
<i>Wires, connectors, pins, crimping pliers and stuff.</i>			
<i>Toolhead of choice</i>	1		
<i>Hotend of choice</i>	1		
<i>Extruder of choice</i>	1		

If you are planning for Medusa HC:

Medusa HC is a toolhead changer specifically designed for the Duender offering by default 4 swappable hotends.

If you use a board of your choice, with for example a Raspberry Pi as your Klipper host, you may have to add a second board to get enough hotend heaters and thermistors ports available.

However, if you are considering using the BTT Manta 5P recommended here, and planning for Medusa HC, it is strongly advised to replace it right away with a **BTT Manta 8P** with the CB2 board and the TMC2209 drivers so you can expand and connect the hotends later on with no added modification.

Voron Duender print settings:

Material: ABS or ASA for all load-bearing parts

You can use a reinforced material (e.g. ABS-CF).
Electronics box and display holder may be printed
in any material

Layer height: 0.2 mm

No. of walls/top/bottom layers: 5 or 6

Infill: > 50% (**60% recommended**) for load bearing parts
Gyroid for strength, **cross-hatch** for speed.
Can be as low as 15% for electronics or screen
boxes

Last minute call

Important tips

Ender 3 disassembly

DO NOT DISMANTLE THE POWER SUPPLY. You will use the PSU along with its cover, the power switch and the wire ending with an XT60 plug.

Deconstruction is easier if you first remove all connections from the mainboard; The connectors are held with hot glue, which you can carefully remove with a pair of pliers.

You can pull out the X axis after removing the top bar, which simplifies further deconstruction. The X motor has a QR code sticker, there are four screws underneath it.

You can easily remove the belts once you've loosened the belt tensioners.

The 4040 central profile and plastic parts won't be needed. Remove the rubber feet for easier squaring, but you may put them back later.

Depending on your batch, you may have to **take the Ender 3 motors pulleys off**. Some use a [printed tool](#), some just Dremel them out. Be careful!

Z Axis

Alternatively to the 3 mm thick aluminium plate that needs to be cut and drilled for the Z axis plates, Openbuild plates can be used, they are more expensive at 2x€2.75, but have all the necessary holes. **NOTE: you must use a [remix of the bed carriages](#) and adapt the screws (flat heads).**

It is probably a good idea, to replace the Z motors shaft couplers with "spring motor coupler" ones at approx. 2x€2.50. In the same domain, it is advised to replace original brass lead screw nuts with anti-backlash spring loaded ones at approx. 2x€2.50 too.

AB Axis

As this guide mentions, the original design suggests to cut one of the existing 2020 extrusions to size (top bar of the Ender 3) for the X axis, and considers an added top front brace to be "optional, but highly recommended".

It is however recommended to **buy a 300 mm 2020 extrusion** for the X axis instead and use the second 330 mm 2020 extrusion for the front. This will add to the rigidity of the frame. Note that MedusaHC project will use a 350 mm 2020 extrusion here.

Toolhead (NOTE: because of Medusa HC, Dragonburner, with a Bambulab X1 series hotend and Sherpa mini extruder are the parts of choice.)

You can adapt many toolheads to the X rail carriage. Most popular choices include XOL, DragonBurner, and even original Ender 3 direct drive mods.

IMPORTANT NOTE: your choice may affect the heatbed positioning. You may need to unscrew lightly base and top Z rails axis M5x8 screws to slide the whole Z assembly front or back a few mm so your printhead travel can cover the whole size of the bed.

Last minute call

Electronics

While you can go with whatever board accepting Klipper, the BTT Manta M5P with a CB1/2 is likely the best choice but you can of course use a Raspberry Zero 2W and re-use the existing boards if you want to be really frugal. **NOTE** that flashing firmware on Creality boards can be tricky.

You may not need a display, as you'll be able to use the web interface anyway, but the original 12864 of your Ender 3 can do the job, or you can get a proper Klipper compatible touch screen. Original design goes with a BTT TFT 35 SPI screen, but it's not as responsive as you'd want.

Remember that you might need different hardware if you made different electronics choices.

Building

During the frame assembly, **it is critical that everything ends up square and flush**. The base and four vertical posts should be placed first then squared while gradually tightening the frame. It is tedious but the print quality of your Duender depends on it.

Since MGN9H rails are about as wide as the V-slots, it's not easy to get them centred. You can [print some spacers](#) or alignment tool like [this](#) or [this](#).

The belt tensioners are also quite fiddly since you need to get all washers and idlers lined up before putting the screw in. [A printed jig](#) may help you.

If you are unsure about how to correctly tap your extrusions, check out this video first:

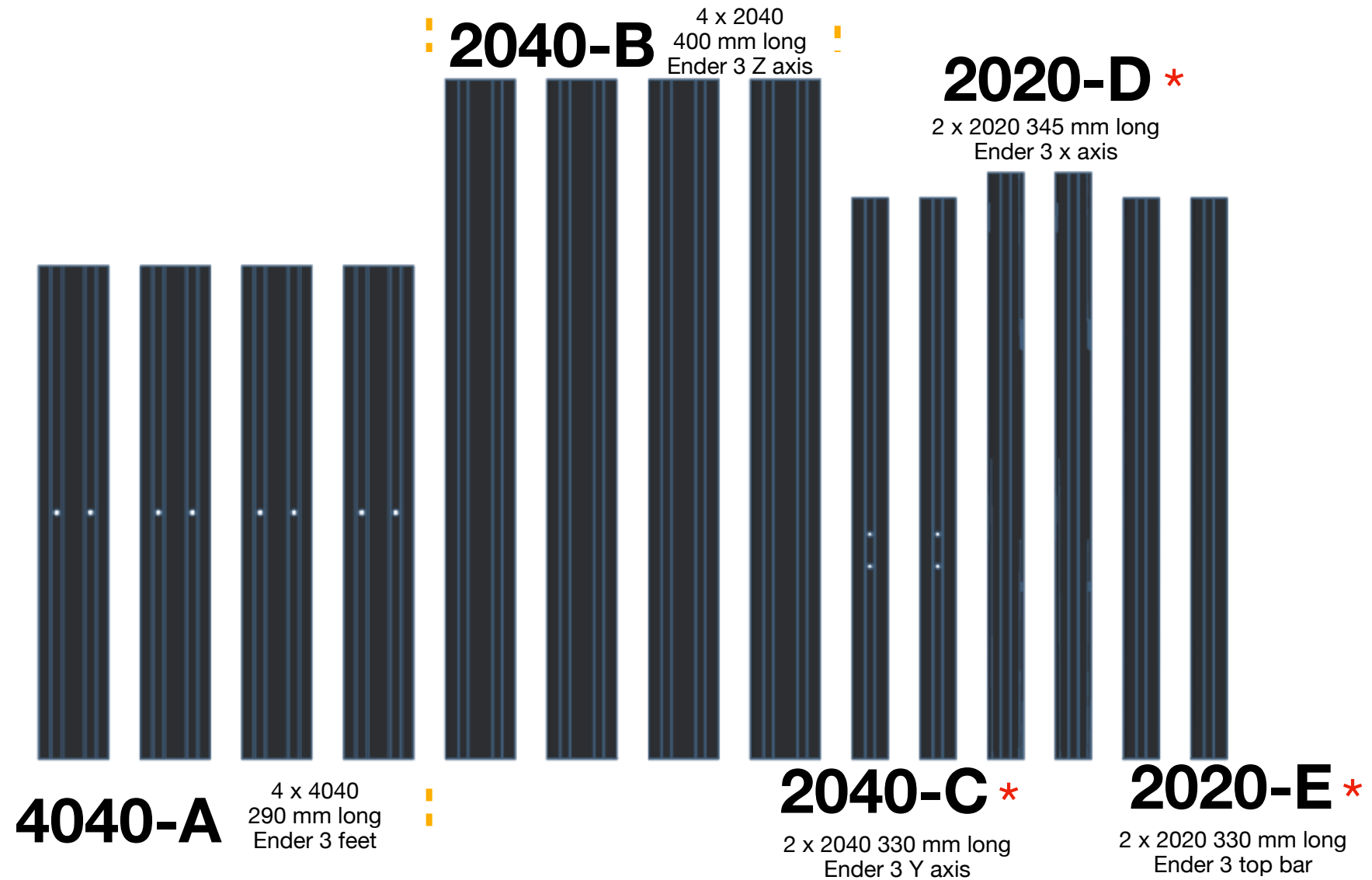


<https://2cm.es/1IYBi>

Preparation

Extrusions reference

6



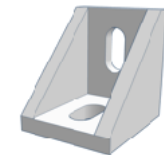
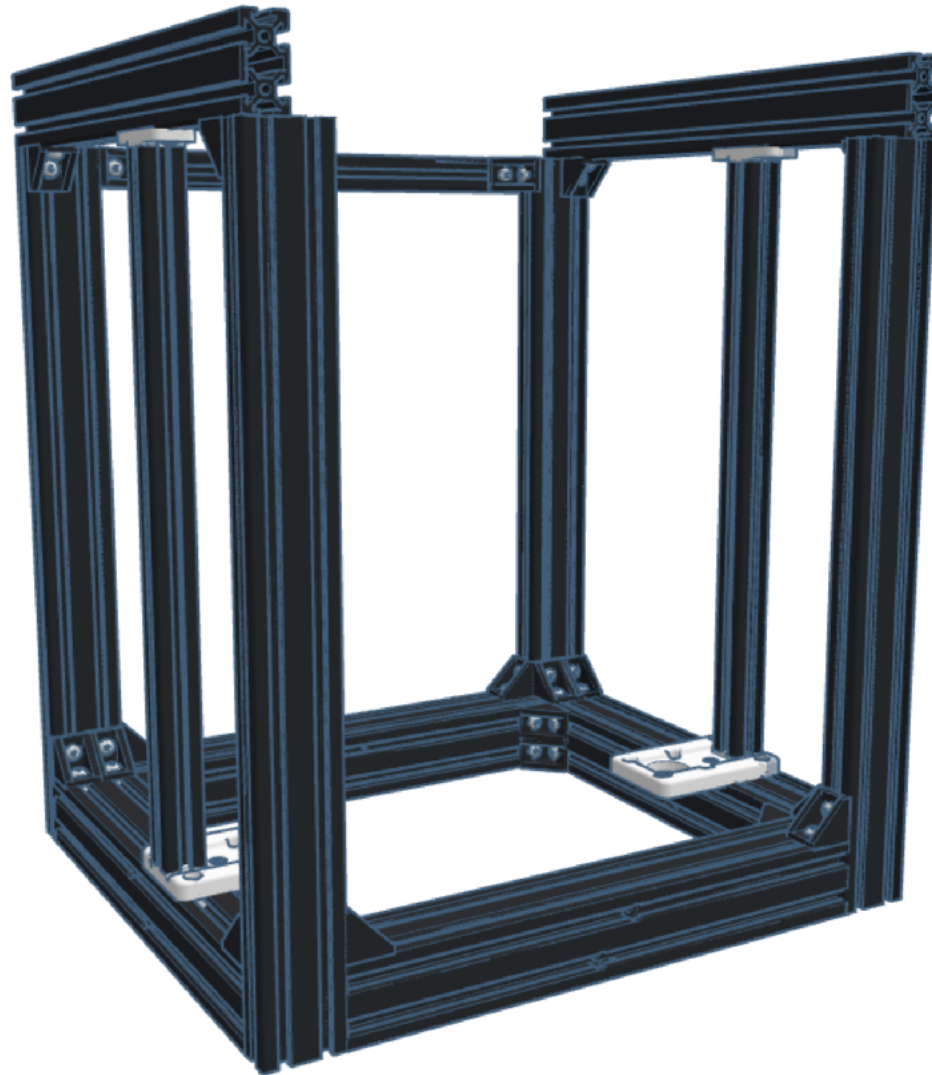
*

Extrusions marked with an * need to be M5 tapped at both ends: 2040-C (8 channels), 2020-D (4 channels) and ONE of the 2020-E (2 channels).

Base structure

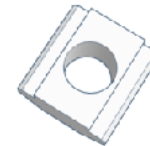
Assembling the frame

7



Metal corner bracket

26 (28)



M5 T-slot nut

69 (71)



M5x8 BHCS

66 (70)

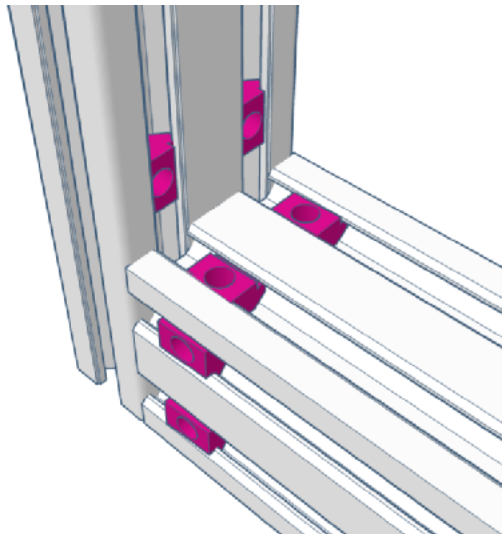
(Figures in brackets are with optional front brace)

Base structure

8

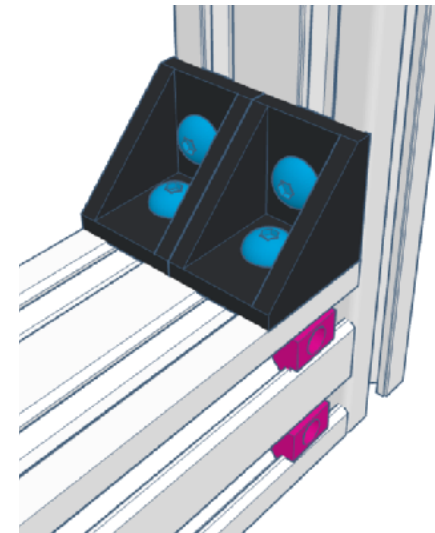


First, prepare each 4040-A with 11 M5 T-slot nut as shown here
You can keep the original Ender 3 plastic pads as feet on the underside
(22 M5 T-slot nuts total)



Insert 2 M5 T-slot nuts in both the 2040-B
(8 M5 T-slot nut total)

**Make sure to orient the back 2040B
extrusions so the mounting holes for the
power supply are towards the inside of
the printer (see picture on next page)**



Place 2 brackets and
insert the M5x8 screws

Repeat at the other end
of each 4040-A

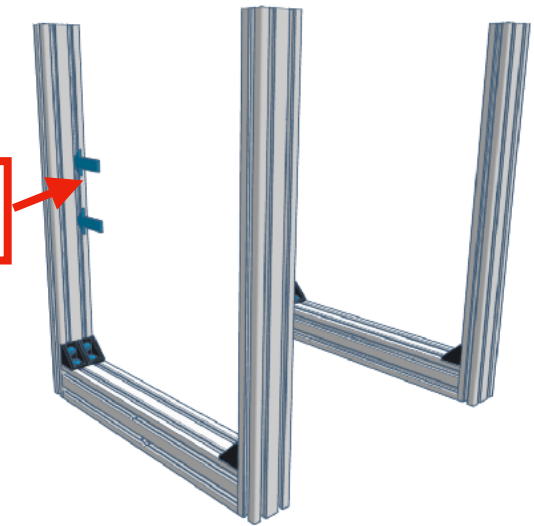
Base structure

9

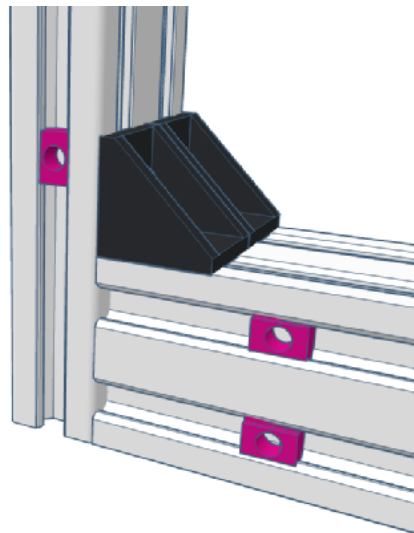


Make sure everything is flush and square before final tightening

Make sure the holes for the power supply end on the inside of the back 2040-B



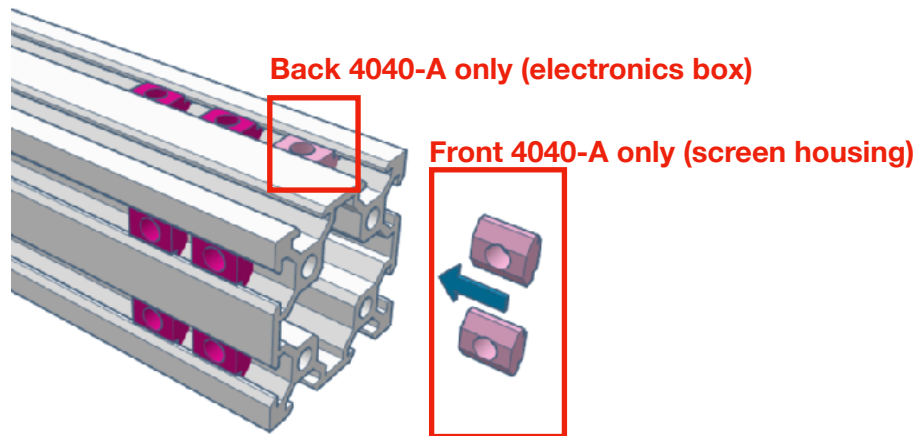
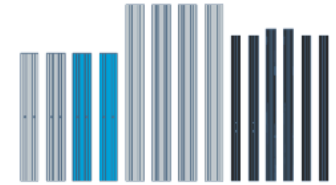
Once done, repeat the same steps with the remaining 4040-A and the two 2040-B to form the opposite side of the frame



Insert a M5 T-slot nut in the inside slot of each 2040-B post of the previous assemblies (4 M5 T-slot nuts total)

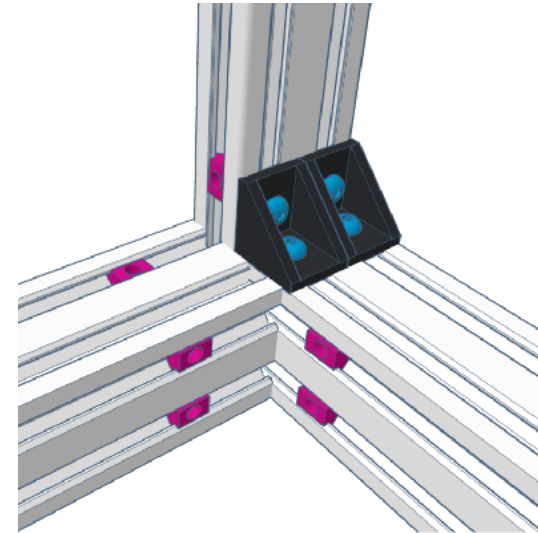
Base structure

10

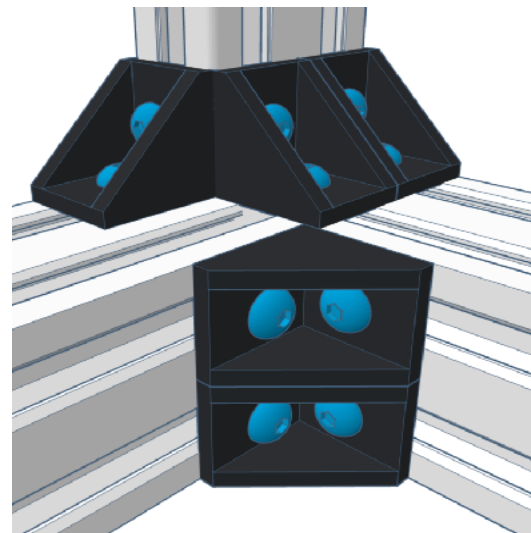


Insert 6 M5 T-slot nut in each of the remaining 4040-A part as shown here (12 total)

Add the M5 T-slot nuts shown in picture in the relevant extrusions (15 total)



Place the 4040-A flush and square to one side assembly and proceed installing the brackets with 6 M5x8 screws on each end (24 total)



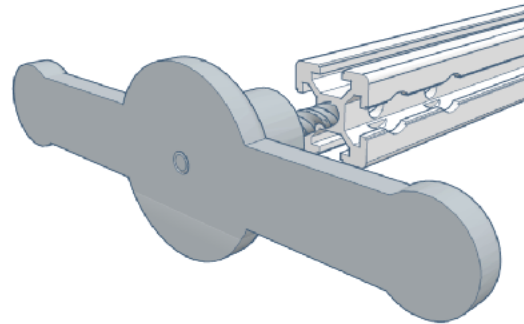
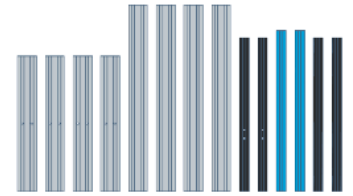
Repeat those steps in the four corners of the base

Once done, double check all your alignments

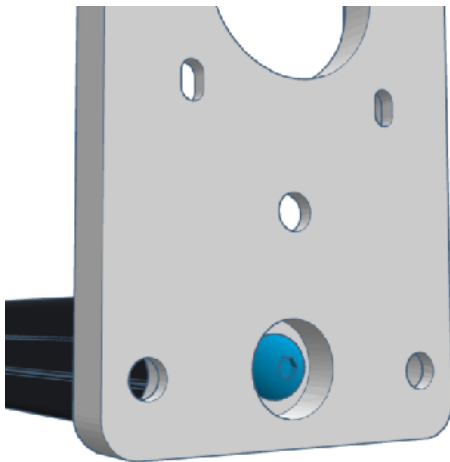
It's a good idea to measure each diagonal of the base square to make sure they are equal too!

Base structure

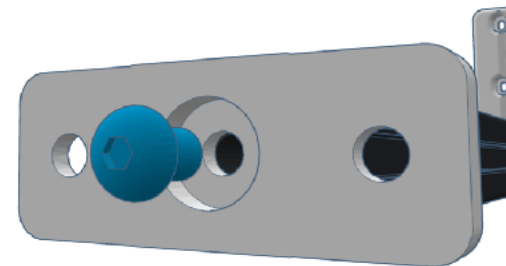
11



Carefully tap each end of the channel in both the 2020-D extrusions with an M5 bit



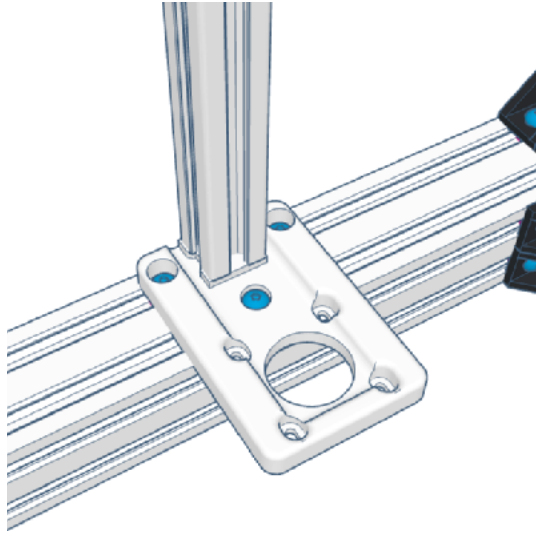
Install the Z motor mount onto one end of the extrusion, making sure it's seated and flush, then secure it with a single M5x8 BHCS (2 M5x8 BHCS total)



Do the same thing with the Z axis top spacer, as pictured (2 M5x8 BHCS total)

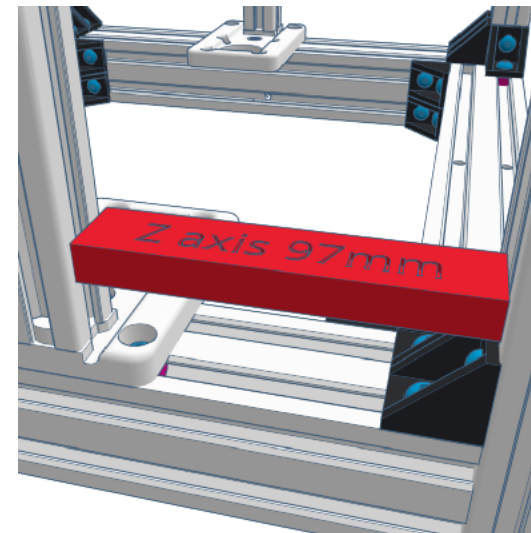
Base structure

12



Install the Z axis onto one side of the base
Align the previously installed M5 T-slot nuts with the holes
Secure it with three M5x8 BHCS
(6 M5x8 BHCS total)

DO NOT FULLY TIGHTEN THE SCREWS YET
The axis needs to be able to slide side to side

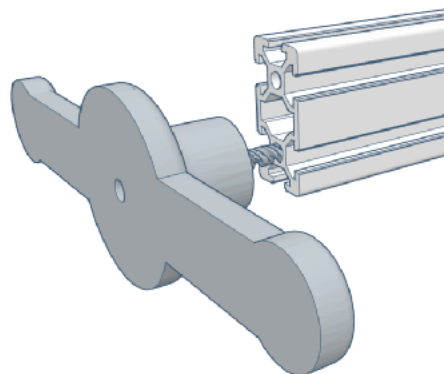
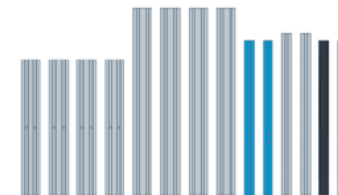


Use the 3D printed spacer provided in the tools
OR a calliper to adjust the Z axis 97 mm from the front post, then tighten the 3 M5x8 BHCS of the base

This is the default distance. You may need to adjust this distance later, depending on your toolhead configuration

Base structure

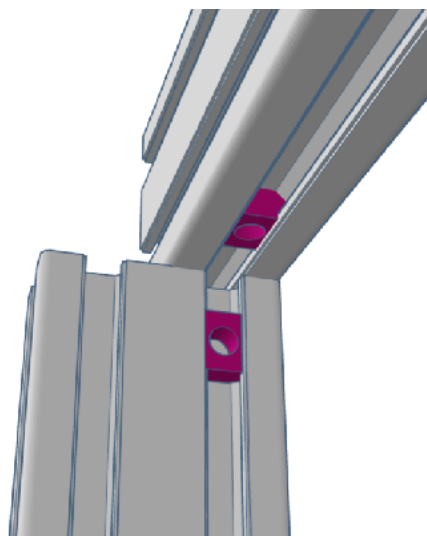
13



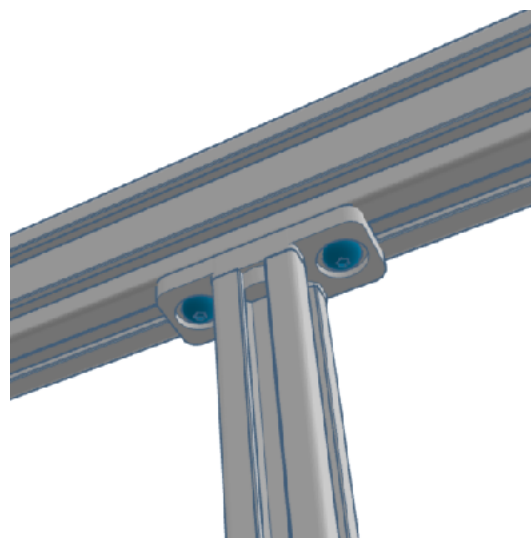
Check if channels of your 2040-C extrusions are already tapped. If not carefully tap each one on each end of both extrusions with an M5 bit (8 channels total).



Once done, insert 4 M5 T-slot nut on the underside of each one (8 M5 T-slot nuts total)

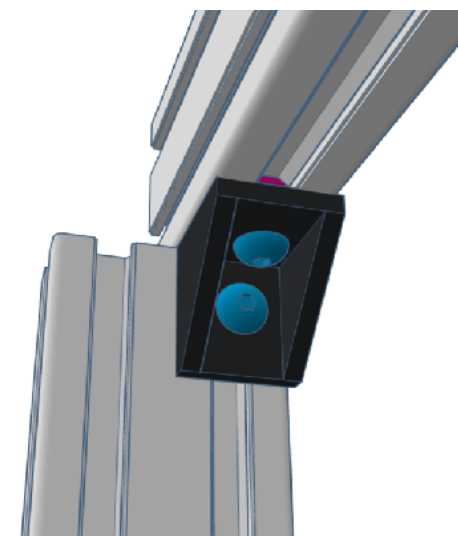


Insert an M5 T-slot nut on the top of each post, in the outside rail as shown (4 total)
Place the 2040-C on top of each side



Align the 2 M5 T-slot nuts remaining under the 2040-C with the top Z axis plate, and use 2 M5x8 BHCS to secure it

Check the distance to the front post to be 97 mm all along the Z axis before tightening the screws

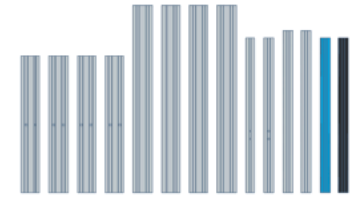
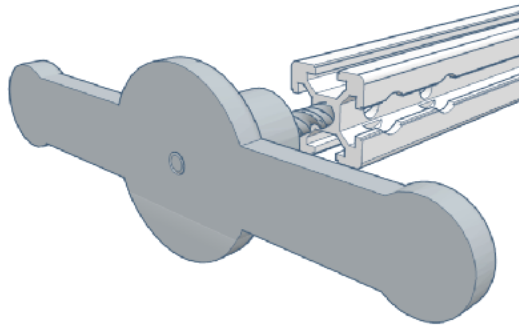


Make sure the extrusions are flush at each end then proceed and install the corner brackets (4 total)

Repeat those steps on the other side to assemble the second Z axis to the frame

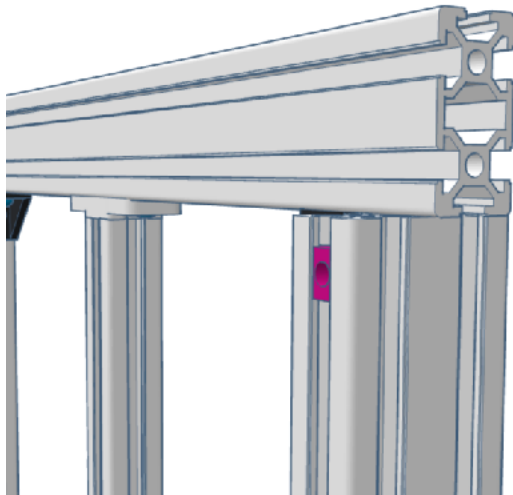
Base structure

14



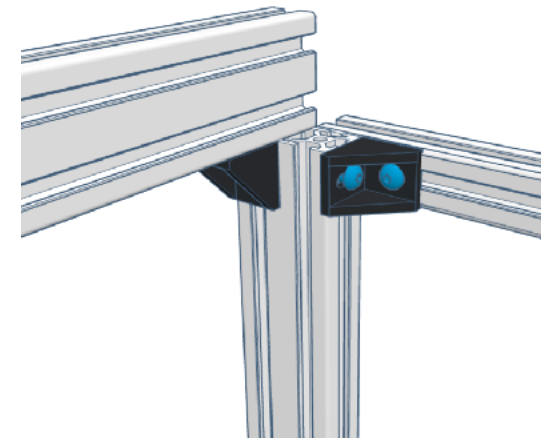
Once done, insert 2 M5 T-slot nuts into the back rail, and 2 more into the front one

Carefully tap each end of the channel in the 2020-E extrusion with an M5 bit



Insert one M5 T-slot nut at the top of each rear Z posts
(2 M5 T-slot nut total)

For the rear right Z post only: add a second M5 T-slot nut
(for the electronics box later on)

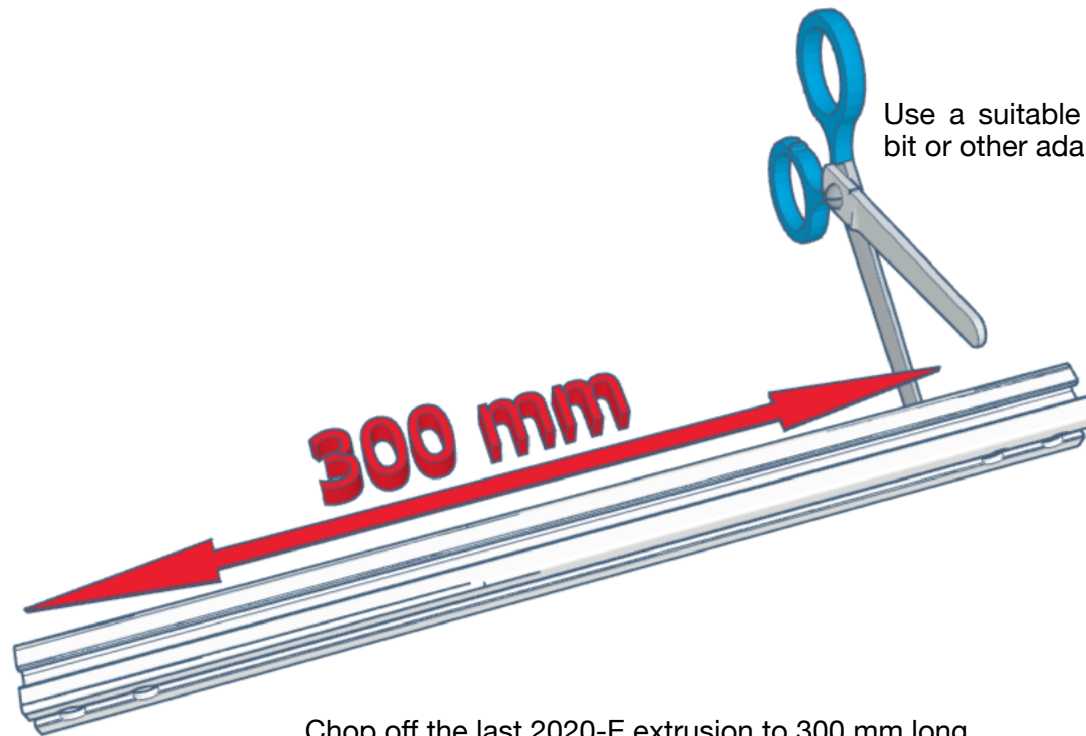
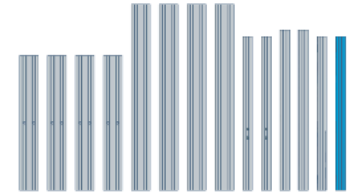


Align the 2020-E extrusion flush to the top of the Z posts

The 2020-E must be centred between the posts

Align the T-slot nuts

Install a corner bracket with 2 M5x8 BHCS
(4 M5x8 BHCS total)



Use a suitable metal cutting saw blade, Dremel bit or other adapted tool to make a clean cut

Chop off the last 2020-E extrusion to 300 mm long and save it to assemble the X axis later on

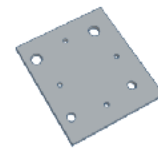
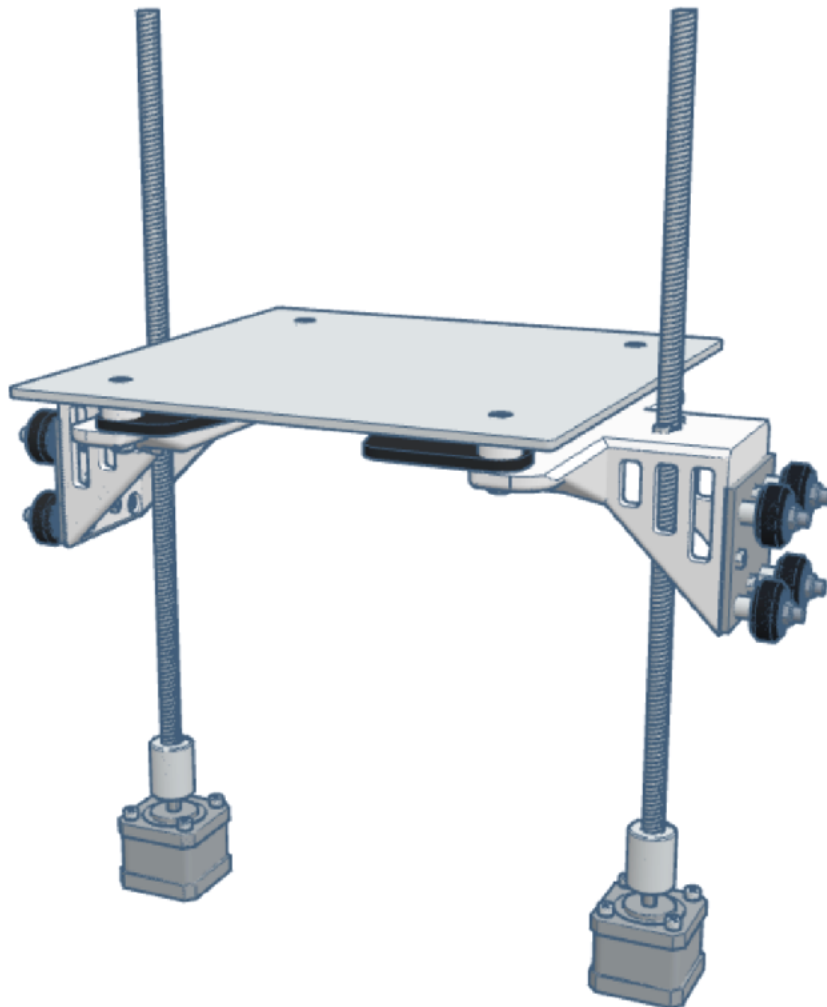
You may prefer to buy a new 300 mm extrusion and keep the 330 mm 2020-E to make a front brace for rigidity purposes.

If that is your case, repeat steps on page 14 for the front part of the printer, and use your newly bought extrusion for the section starting on page 28

NOTE that if you are planning for MedusaHC mod later, the front brace needs to be 350 mm long. So best solution is to still cut 330 to 300 mm and buy a single 350 mm 2020 extrusion for the front brace.

Assembling the Z axis Z gantry

16



DIY aluminum plates
or Openbuild plates



M3x16 SHCS



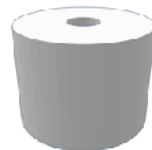
M3x8 SHCS



M4 Nylstop nut



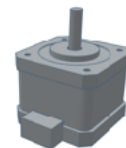
M3 Nylstop nut



12 mm silicone spacers



M4x35 FHCS
(from Ender 3)



Nema 17 42-34
(from Ender 3)

2



M5x30 BHCS
(from Ender 3)

8

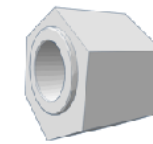
4



M5 Nylstop nut
(from Ender 3)

8

16



Eccentric M5 nut
(from Ender 3)

4

4



Wheel M5 spacer
(from Ender 3)

4

12



POM V-Wheel
(from Ender 3)

8

4



Lead screw nut
(from Ender 3)

2

4



8 mm lead screw
(from Ender 3)

2

2

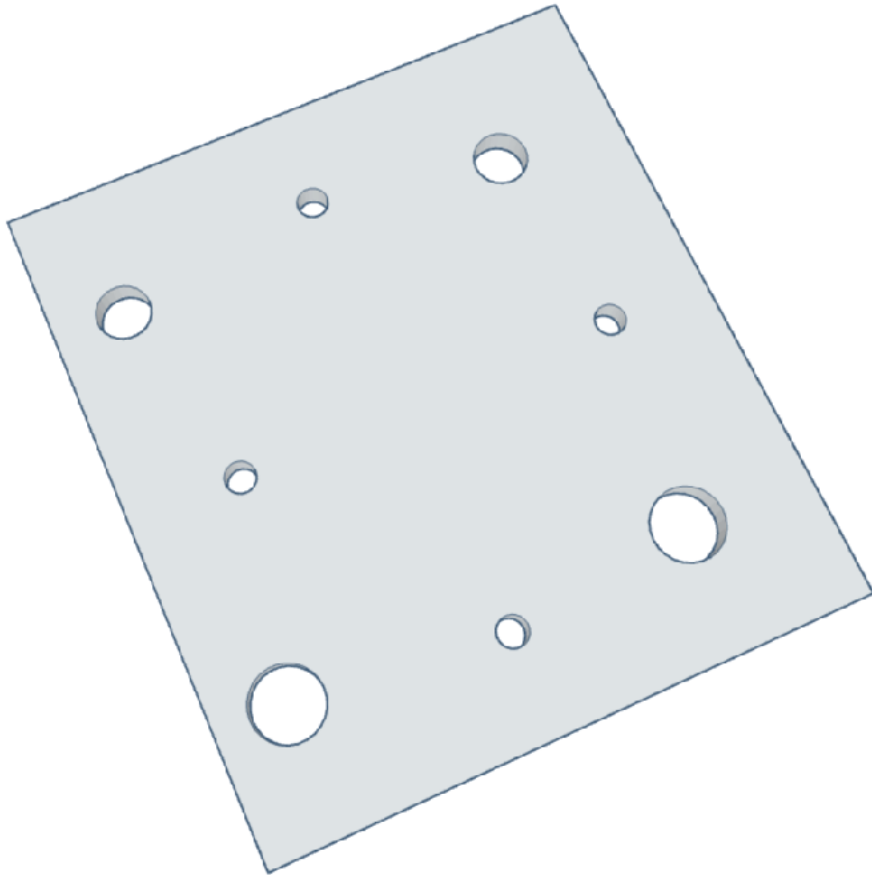


Motor shaft coupler
(from Ender 3)

2

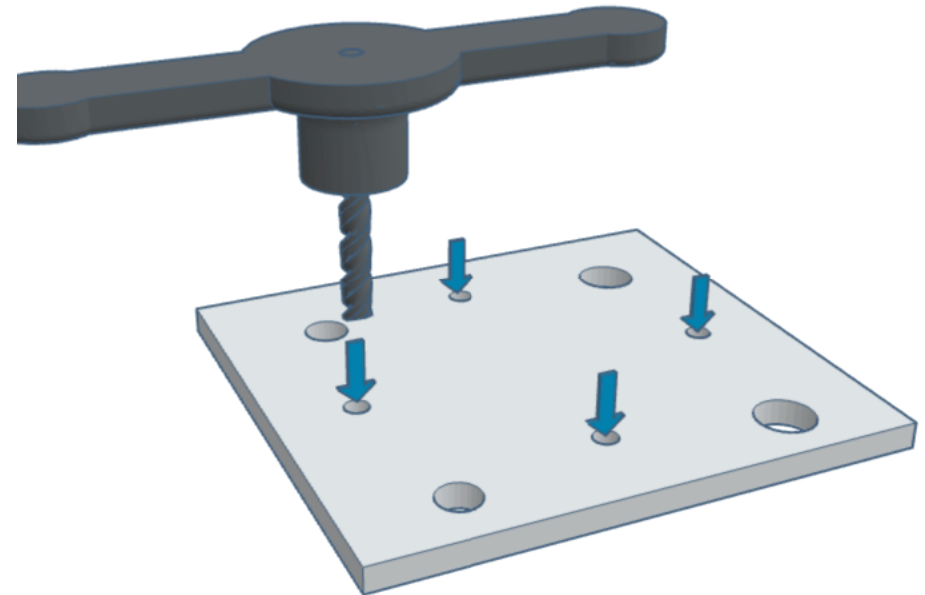
Assembling the Z axis

17



You will need to prepare two plates in 3 mm thick aluminium. The STL templates will give you exact dimensions and holes sizes/locations.

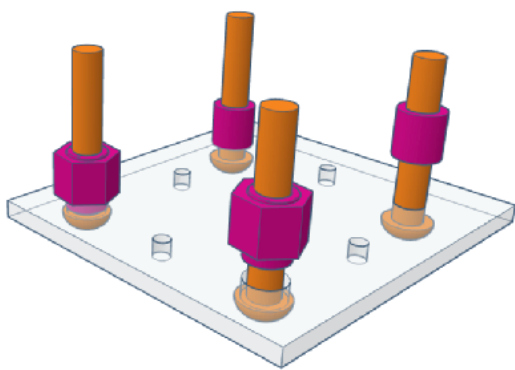
If making those is a problem, a mod using Open build plates exists, with all the necessary holes pre-drilled.



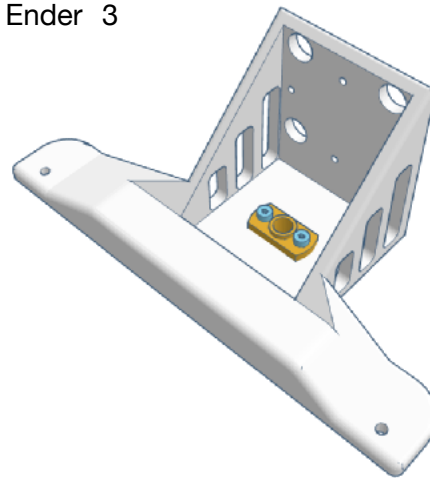
Carefully tap the marked holes with a M3 threading bit

Assembling the Z axis

18

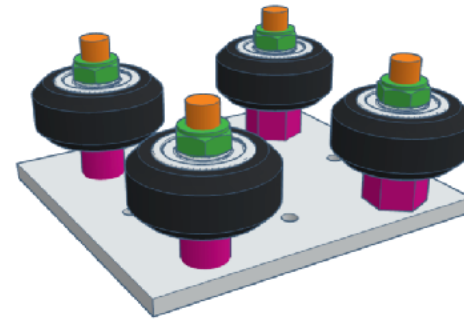


Insert 4 M5x30 BHCS from the underside
Spacers go over the smaller holes and
eccentric nuts over the larger ones
Those are all from the original Ender 3
rollers mounts

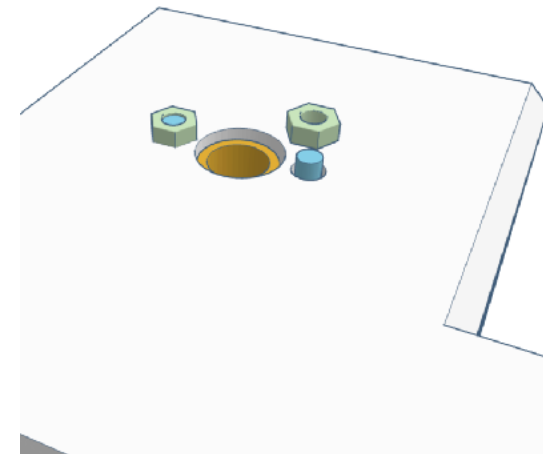


Install the lead screw nuts under the plastic bed holders,
and insert 2 M3x16 SHCS or BHCS (Or you can use the
ones originally coming from your Ender 3)

**If your lead screws aren't perfectly straight, it is
advised that you swap the original lead screw nuts
from the Ender 3 with some anti-backlash ones, to
avoid Z wobble**



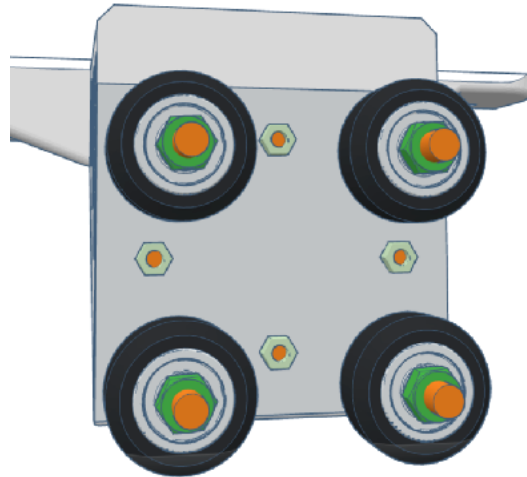
Finally, add 4 V-rollers from your Ender 3, and
an M5 Nylstop (self-locking) nut
**DO NOT TIGHTEN THE NUTS YET, leave
the wheel assemblies loose for adjustment**



Turn the bed holders around being careful not to
lose the screws and secure them with M3 nuts

Assembling the Z axis

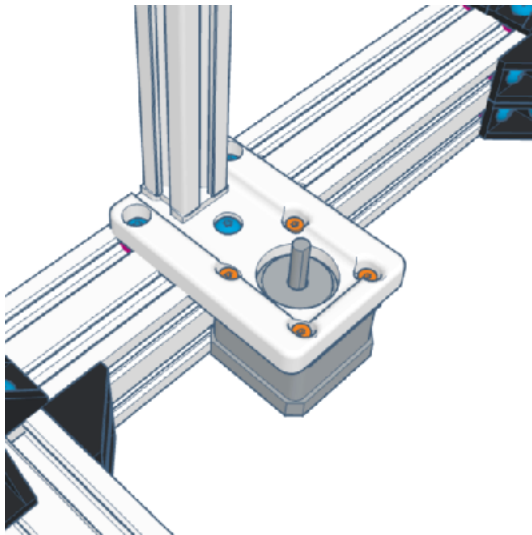
19



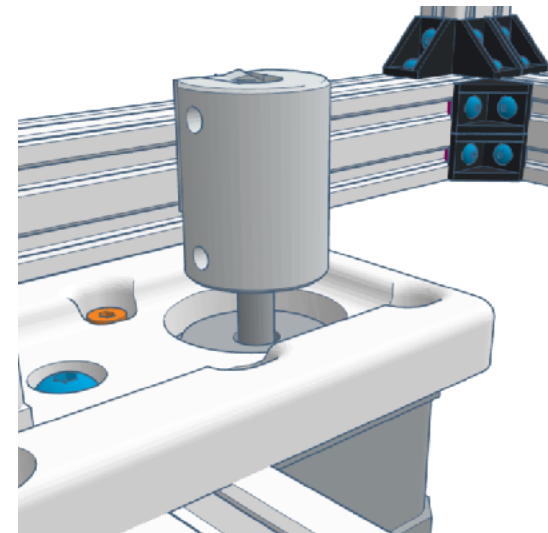
Place the rollers plate behind the bed holder and use 4 M3x8 SHCS to secure it. The screws should bite directly into the threads in the aluminium plate.

Alternatively, you may use M3x12 SHCS and secure them with 4 M3 hex nuts from the back of the plate, as pictured.

Remember that if you used open builds plates, screws will go back to front, and you'll want to double check for their head not touching the Z extrusion.



Place the Z motors under their plates and use 4 M3x8 SHCS to secure them. Orient the motor plug to the back of the printer

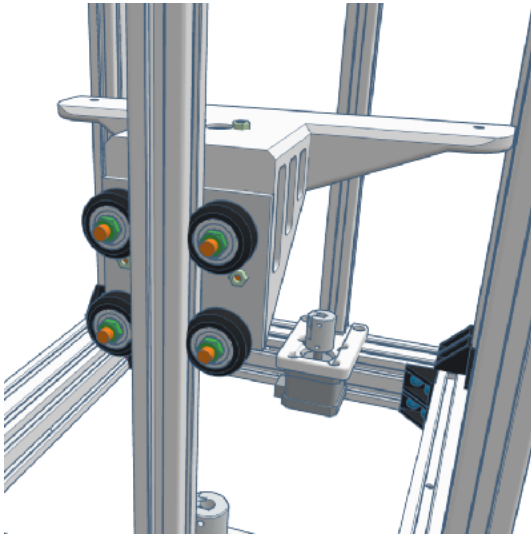


Install the lead screws couplers onto the Z motors shafts and tighten only the lower screw

If your lead screws aren't perfectly straight, you may want to replace those with "spring" or "elastic" ones to avoid Z-wobble

Assembling the Z axis

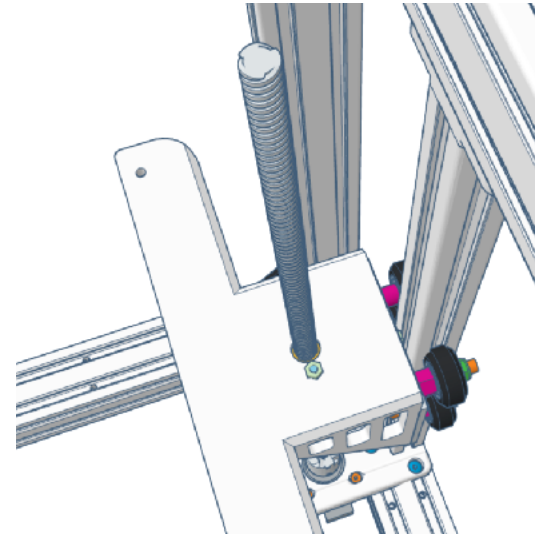
20



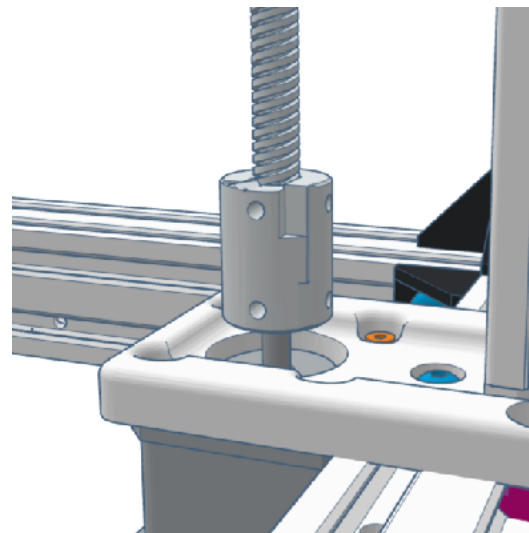
Insert the bed holder assembly onto the Z axis (note that the longer side goes towards the front) and tighten the V-wheels. Once secured, adjust the eccentric nuts to ensure all 4 wheels apply equal pressure.

The wheels should rotate smoothly with a slight resistance when turned by hand

Double-check that the screw heads don't touch the extrusion surface.



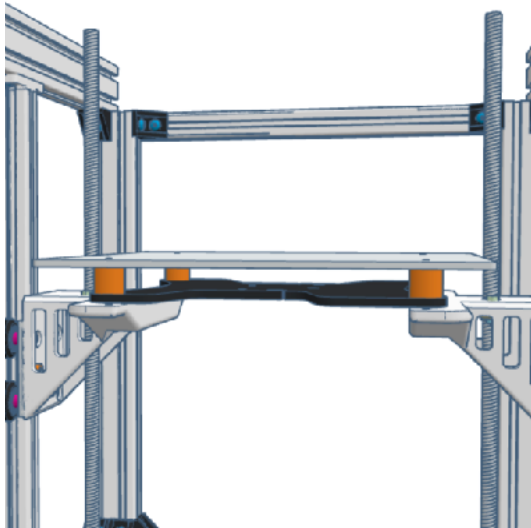
Insert each lead screw into its lead screw nut from the top then insert them into the Z motor coupler by rotating them until fully seated.



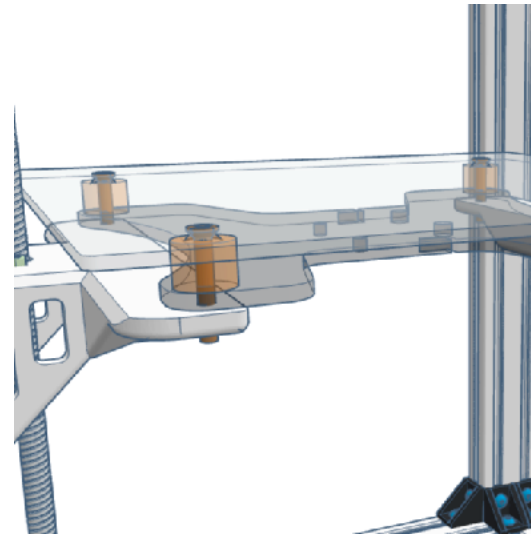
Tighten the upper screw of the coupler to hold the lead screw.

Assembling the Z axis

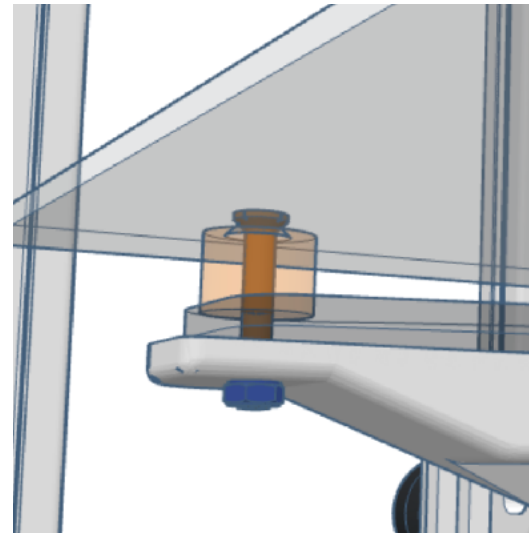
21



Place the Ender 3 Y plate onto the bed holders, followed by 4 silicone spacers (12 mm version shown here in orange) or you may use the springs from the Ender 3, then the heatbed. Of course, the wiring of the heatbed should end towards the back of your printer.



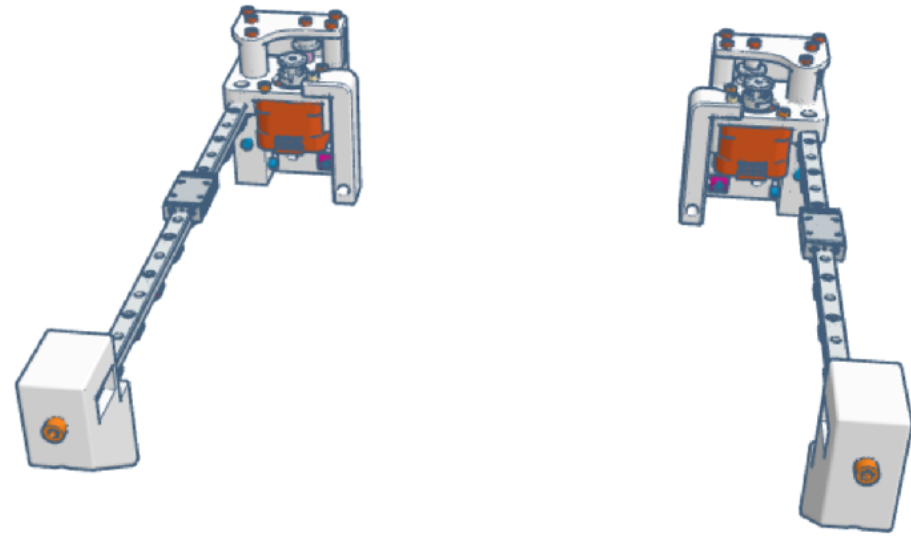
Use the original 4 M4x35 FHCS and secure them under the bed holders with M4 nuts.

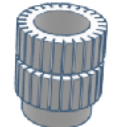
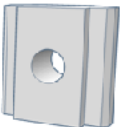
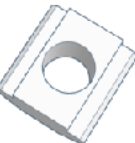
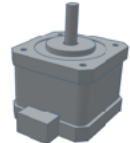


Building the AB kinematics

AB gantry

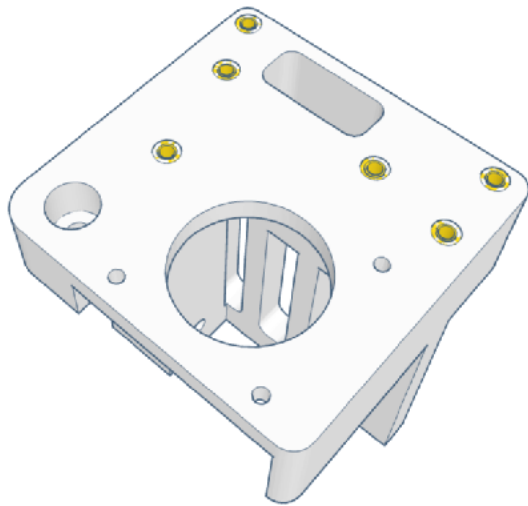
22



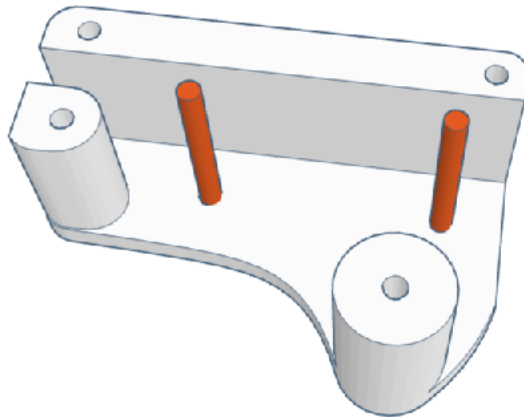
	M3x30 SHCS	12		M3x0.5 spacer	28		MGN9H 300 mm rail	2
	M3x20 SHCS	2		M3x10 spacer	1		GT2 16T6W3B toothless idler	8
	M3x16 SHCS	2		M3 heat-set insert	10		GT2 16T6W5B motor pulley	2
	M3x8 SHCS	22		M5x8 BHCS	14		M5 Nylstop nut (from Ender 3)	2
	M3 T-slot nut	18		M5 T-slot nut	2		Nema 17 42-40 (from Ender 3)	2

Building the AB kinematics

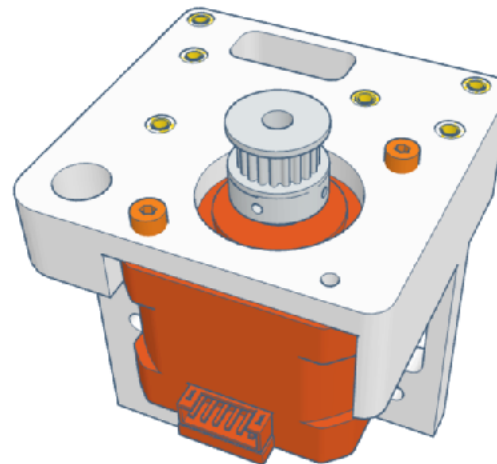
23



Prepare the bottom part of AB motor holders:
with 6 M3 heat-set inserts each side
(12 M3 heat-set inserts total).



Pass 2 M3x30 SHCS into the idler holes and turn both top halves upside down.



Install the motors under the plates with 2 M3x8 SHCS in the shown locations

Mind the motor plug orientation

Also place the motor pulleys onto the shafts of A & B motors

Do not tighten them too much. They will need to be adjusted later, after belts installation

For the Left AB mount:

On the innermost pin (right side when upside down), stack:

- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers.

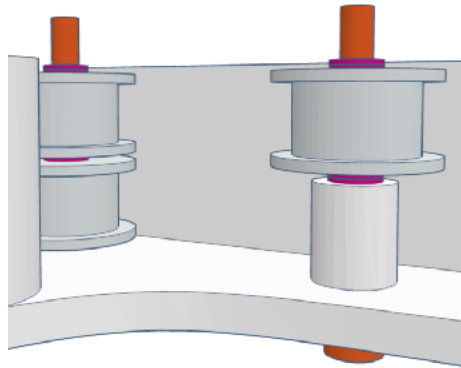
- On the innermost pin, stack:
 - 2 M3x0.5 mm spacers,
 - 1 GT2 16T toothless idler,
 - 2 M3x0.5 mm spacers,
 - 1 M3x10 mm spacer.

The best option is certainly to buy and install a metal M3x10 spacer, but should you not have one on hand, you may use the printed one provided in the stl files.

Building the AB kinematics

24

For the Right AB mount:

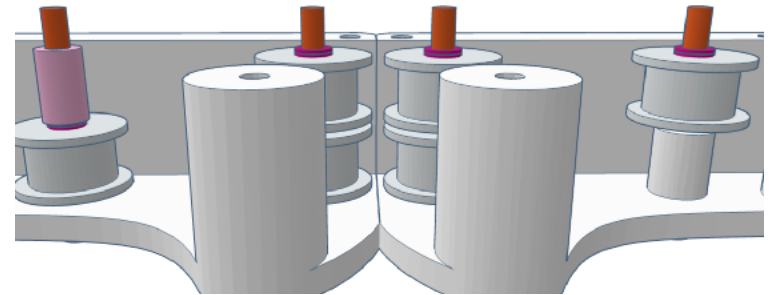


Onto the outermost pin (left side when upside down), stack:

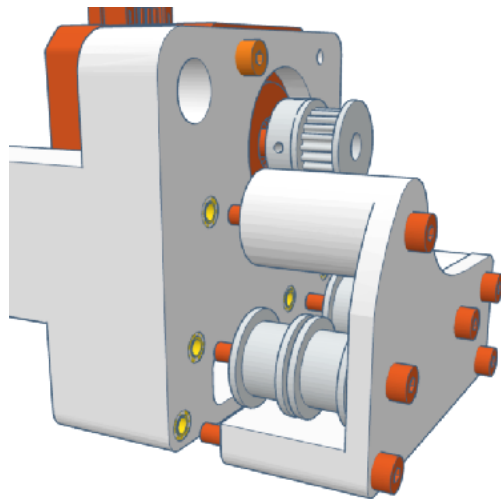
- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers.

Onto the innermost printed post pin:

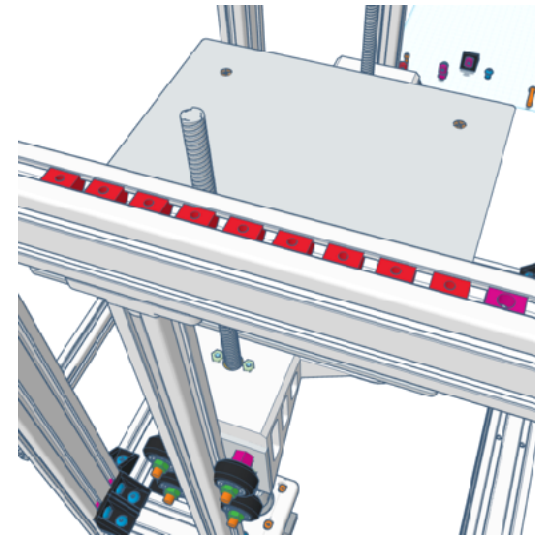
- 3 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers.



When the mounts are placed against each other, the idlers should be perfectly level between both sides. If that is not the case, adjust the spacer count as needed.



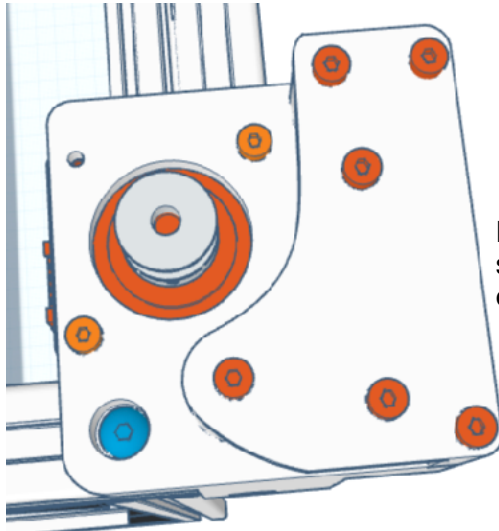
Lay the assembly on its back to prevent the spacers from falling out and assemble the two halves together.
Secure the assembly with 4 M3x30 SHCS on each side.
(12 M3x30 SHCS total)



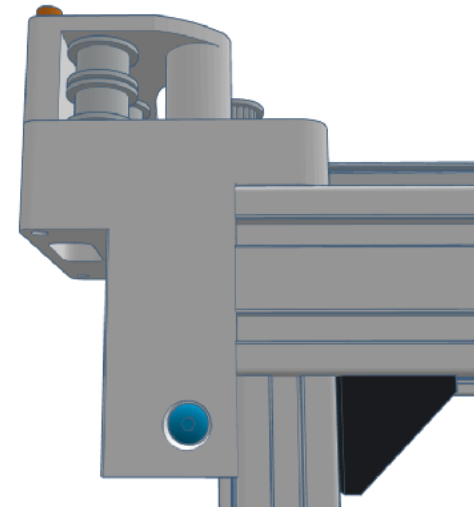
Install 9 M3 T-slot nuts then 1 M5 T-slot nut on top of each Y axis.
If you have hammerhead nuts on hand, it's better to insert only 8 M3 T-slot nuts.

Building the AB kinematics

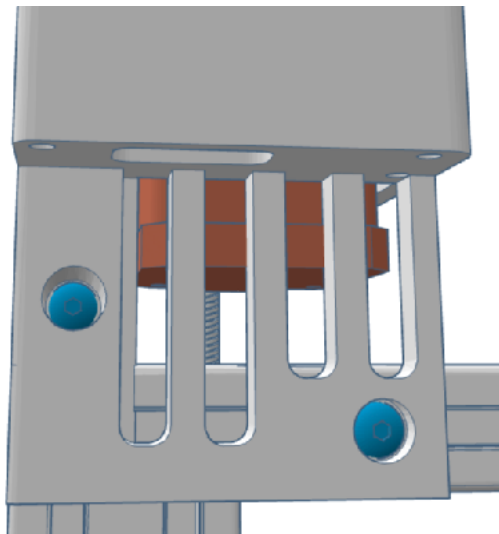
25



Install the AB motor holders on each side and use a M5x8 BHCS to secure it onto the top M5 T-slot nut



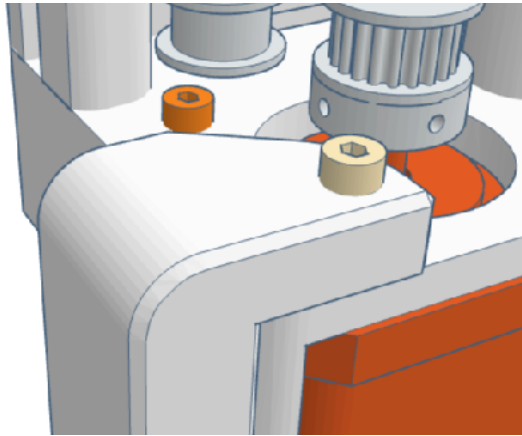
Use a second M5x8 BHCS threaded into the tapped end of the 2020-E back brace



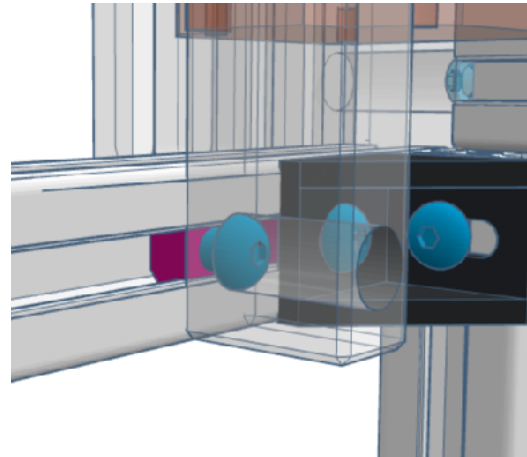
Use a M5x8 BHCS threaded into the tapped end of the 2040-C Y axis and another one into the back brace M5 T-slot nut

Building the AB kinematics

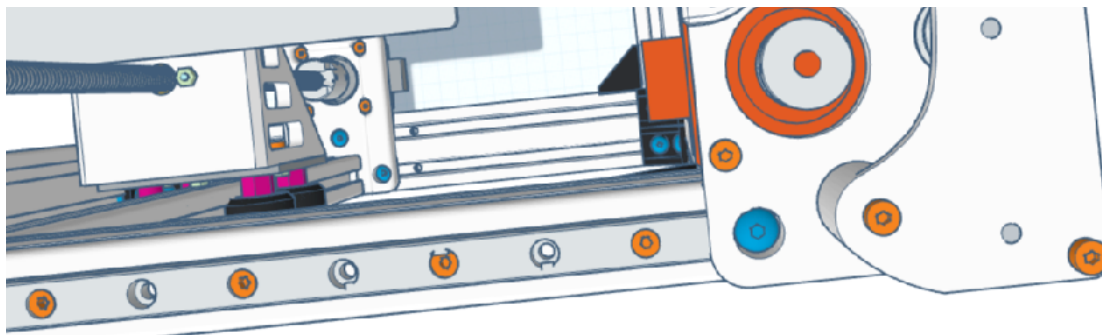
26



Install the motor holders with M3x16 SHCS threaded directly into the motor

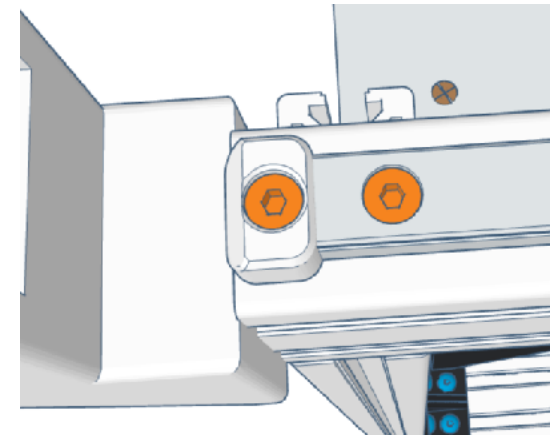


Secure the base of the holder with an M5x8 BHCS into one of the previously inserted T-slot nuts in front of the back brace



Install the MGN9H rails against the motor holders with the help of 3D printed centring jig and 8 M3x8 SHCS

DO NOT FULLY TIGHTEN THE SCREWS YET

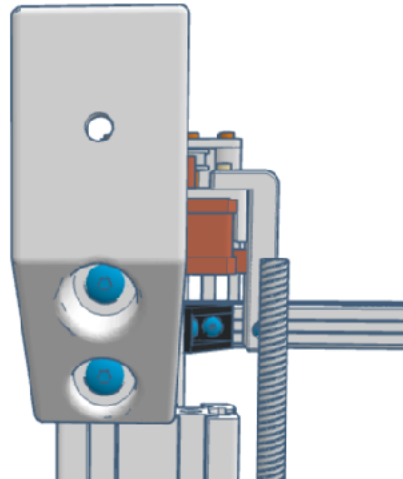


Install the Y carriage stopper with 1 M3x8 SHCS onto the last M3 T-slot nut.

If you have hammerhead nuts, use 2 of them to install the stopper, as it may help you to replace it with another accessory later.

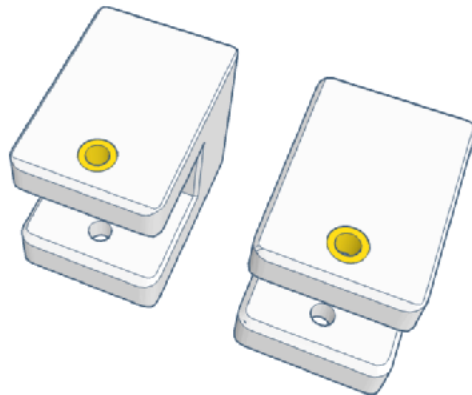
Building the AB kinematics

27

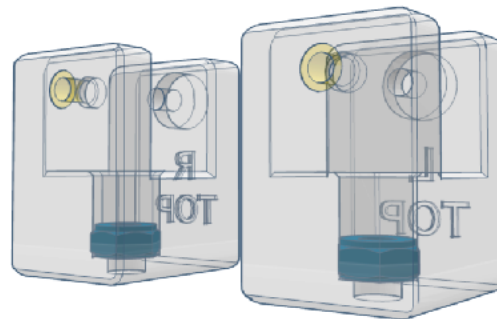


Install the front holders with 2 M5x8 BHCS on each side

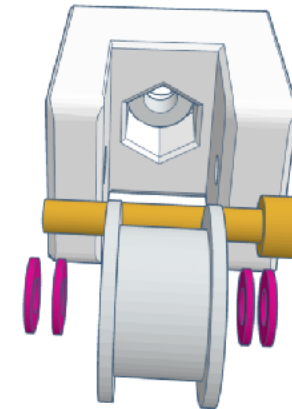
Mind the side, the holders are not symmetrical



Prepare both the Left and Right belt tensioners by placing an M3 heat-set insert in the underside hole



Insert an M5 Nylstop nut into the groove inside both holders. A light dab of glue may be used to ensure they stay in place.



Place

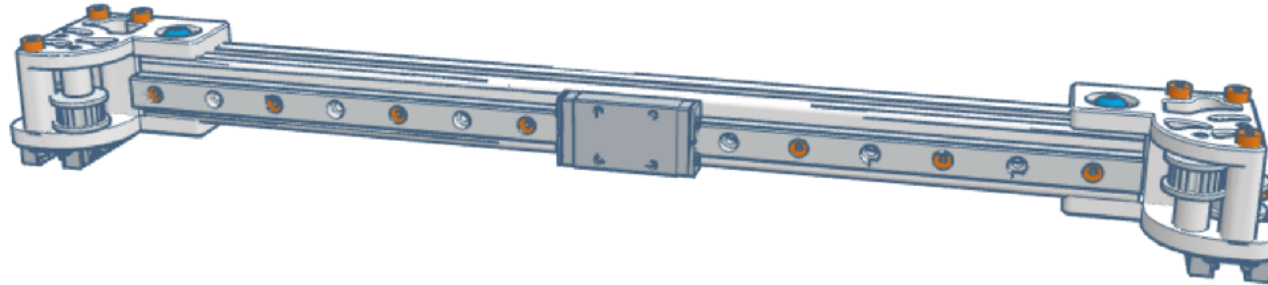
- 2 M3x0.5 mm spacers,
 - 1 GT2 16T toothless idler,
 - 2 M3x0.5 mm spacers,
- all held by a M3x20 SHCS inside each tensioner

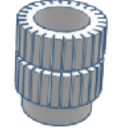
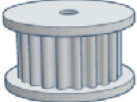
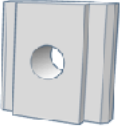
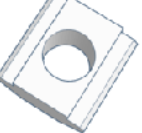
As already mentioned, [a printed jig](#) can make this much easier

Building the X axis

X axis gantry

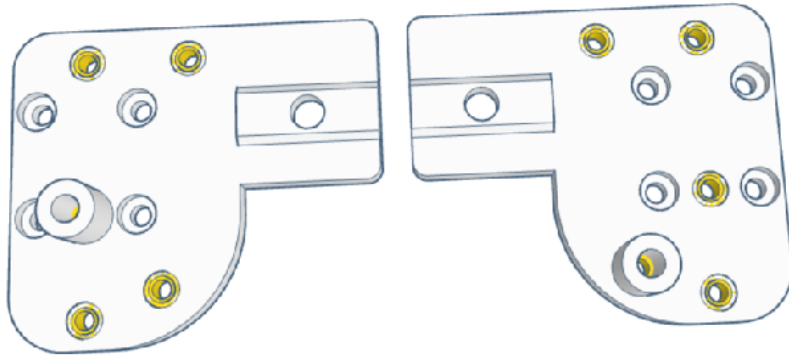
28



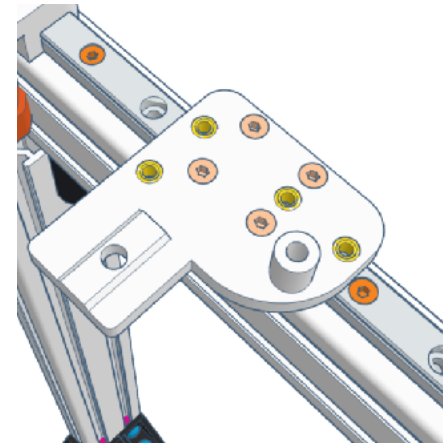
	M3x30 SHCS	10		M3x0.5 spacer	20		MGN9H 300 mm rail	1
	M3x8 SHCS	8		M3 heat-set insert	10		GT2 16T6W3B toothless idler	2
	M3x6 SHCS	8		M5x8 BHCS	6		GT2 16T6W3B toothed idler	2
	M3 T-slot nut	8		M5 T-slot nut	6			

Building the X axis

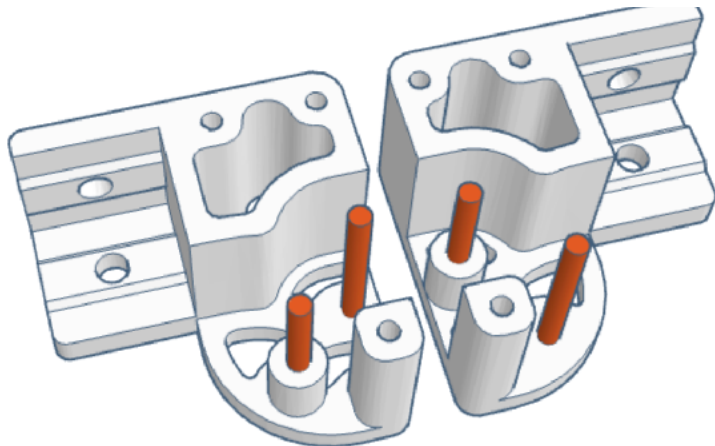
29



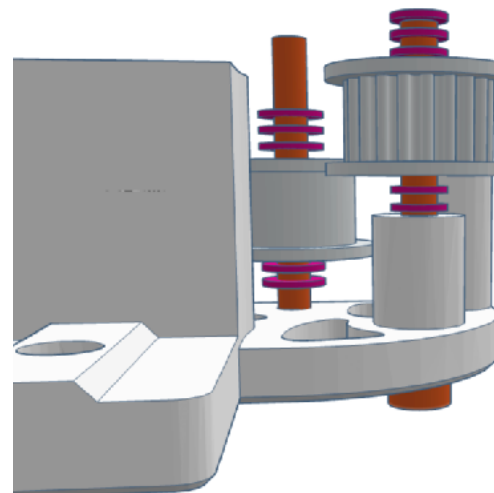
Prepare both bottom parts of the X carriage with 5 M3 heat-set inserts each side.
The inserts in the printed posts are easier to install from below.



Attach both bottom halves onto their rail carriage using 4 M3x6 SHCS (8 M3x6 SHCS total).



Turn both top halves upside down, and insert 2 M3x30 SHCS each side to serve as the idler axis



Each half takes:

Onto the frontmost axis

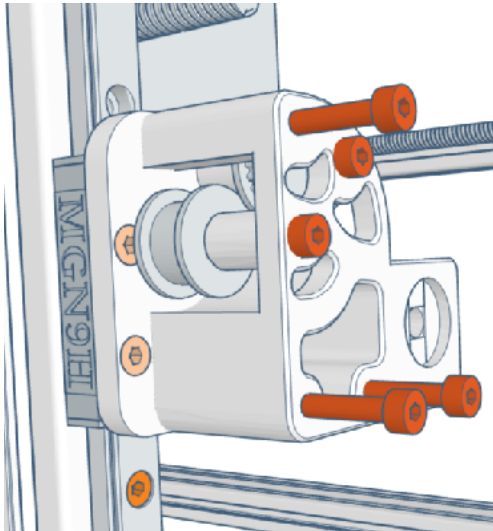
- 2 M3x0.5 mm spacers,
- 1 GT2 16T **toothed** idler,
- 3 M3x0.5 mm spacers.

Onto the rearmost axis:

- 2 M3x0.5 mm spacers,
- 1 GT2 16T **toothless** idler
- 3 M3x0.5 mm spacers.

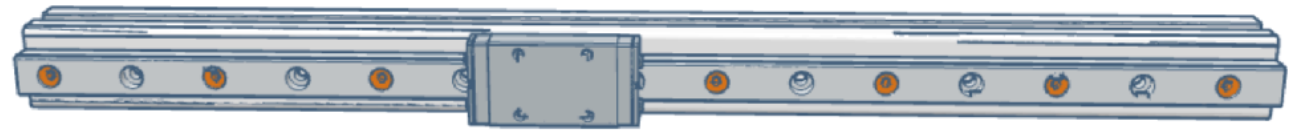
Building the X axis

30

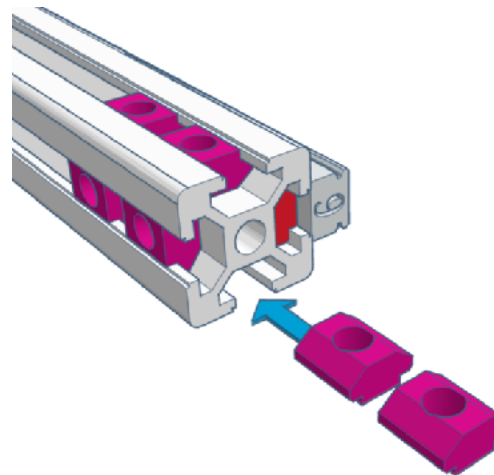


Lay the printer on its back to prevent the spacers from falling out and tighten both sides of the X axis assembly, adding 3 M3x30 SHCS, to assemble the two halves.
(6 M3x30 SHCS total)

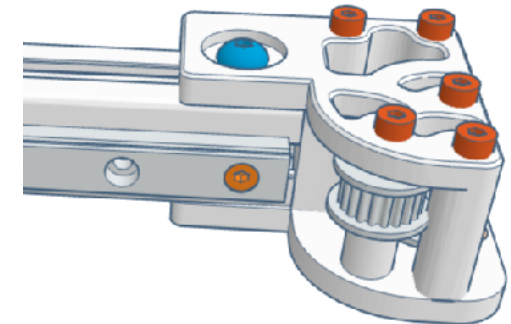
Just screw enough so that parts hold together: both halves need to be loose for the next step.



Assemble the 8 M3 T-slot nuts to the last 300 mm rail with 8 M3x8 SHCS. Leave them loose enough to slide into the 300 mm X axis extrusion.
Center the rail flush both sides and, using a centring jig, tighten the screws to secure the assembly.
(8 M3 T-slot nuts and 8 M3x8 SHCS total)



Insert 2 M5 T-slot nuts on the top side, 2 on the back side and 2 more under the X axis extrusion
(6 M5 T-slot nuts total)

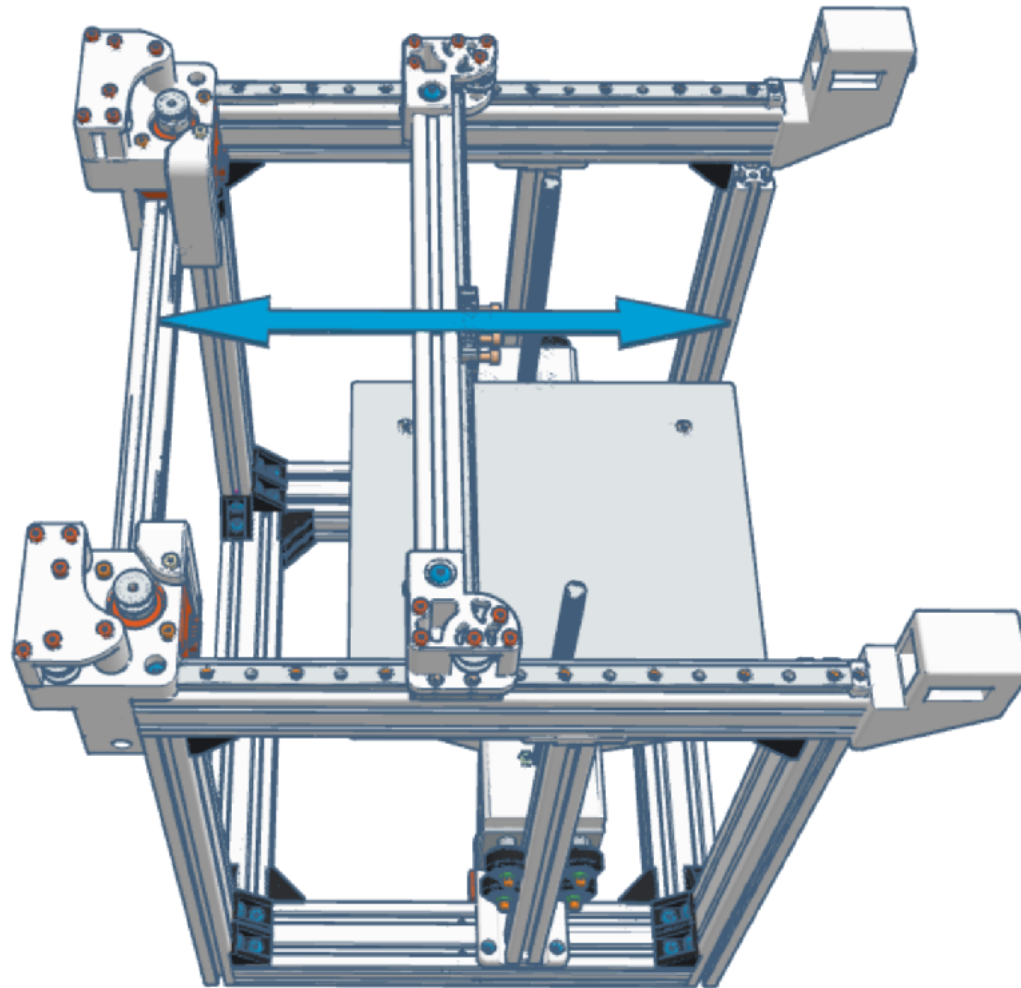


Push the X axis into its carriages on the two sides of the printer, carefully centre it, then tighten both carriages.

Once done, use the previously inserted T-nuts to tighten 3 M5x8 BHCS at each end:
1 on top, 1 on the back and 1 under the X axis.
(6 M5x8 BHCS total)

Building the X axis

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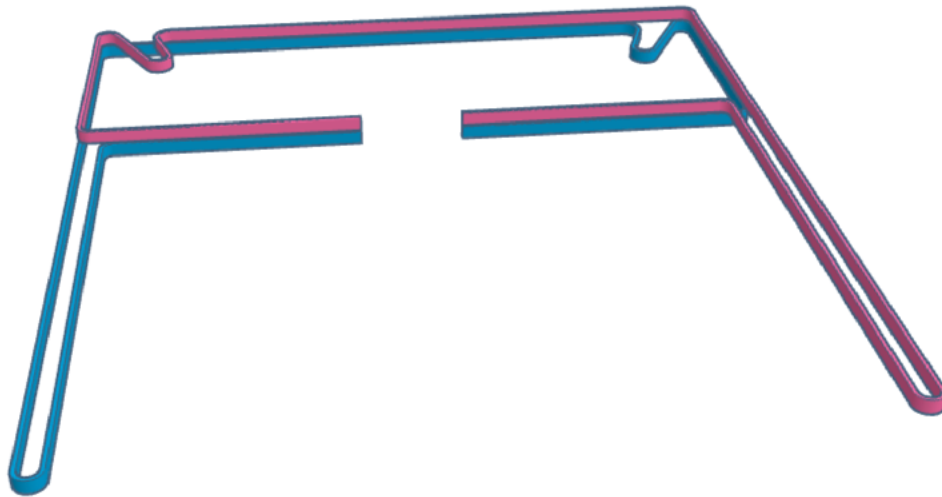


Finally, slide the X axis assembly back and forth several times to allow the Y rails to self-align. Gradually tighten the M3x8 screws each side from one end to another to lock the rails in place.

(Note that the toolhead carriage will be attached to the rail with 4 M3x6 SHCS shown here.)

A & B belts installation GT2 belts routing

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GT2 6 mm belt approx. 2x1,5m



M5x25 SHCS
(from Ender 3)

2

"Equal belt tension is important to the proper function of a CoreXY motion system."

We recommend to run one belt to get the required length, remove the belt from the printer and cut the second belt to the exact same length.

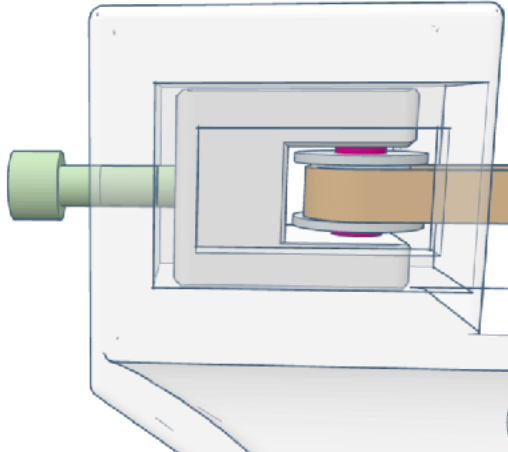
As both belt paths have the same length this is an easy way of getting a consistent tension."

- The Voron team

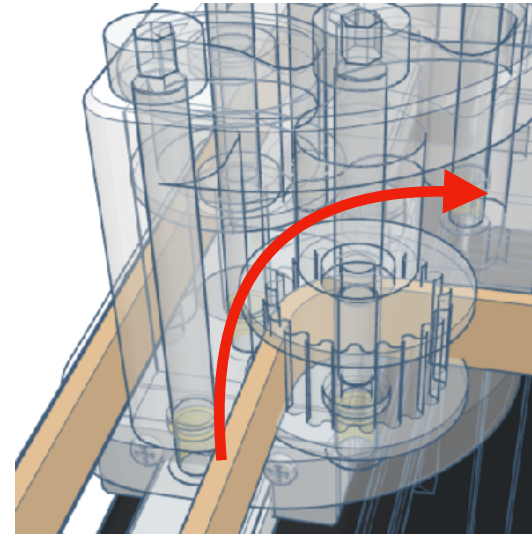
A & B belts installation

33

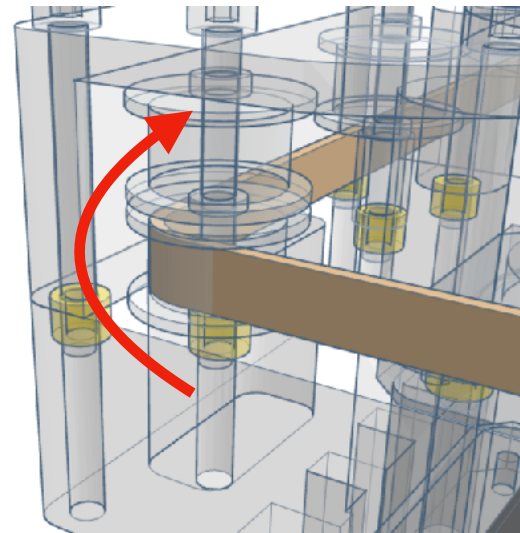
A belt



Pass the belt into the L side tensioner (flat side against idler) then insert the tensioner into the L front holder, and secure it with a M5x25 SHCS passing from the front of the holder



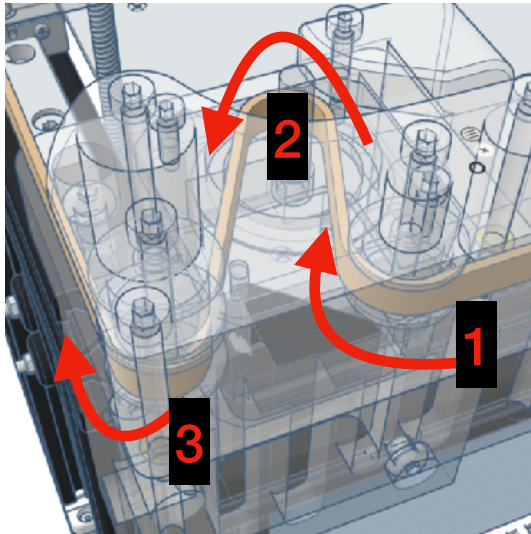
Pass the short side of the belt into the toothed idler and towards the X axis toolhead carriage



The longer side passes next to the left Y carriage all the way to the AB motor holder at the back left
Use the lower idler to turn 90° and head to the right AB motor holder

A & B belts installation

34



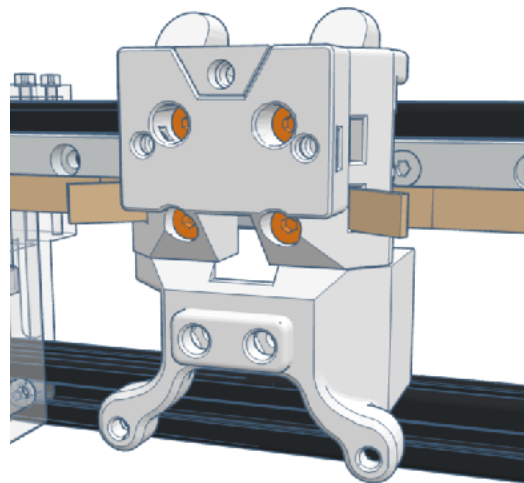
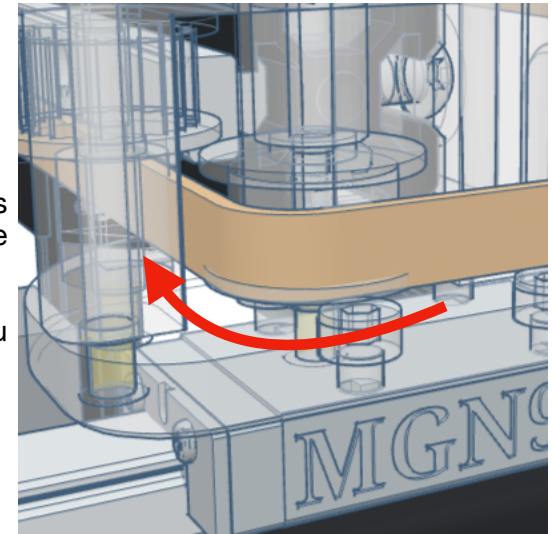
1: Into the right AB motor holder, use the first idler to redirect the belt towards the motor pulley (not shown here)

2: Place the belt into the A motor pulley, **it's time to adjust pulley height.** Then back to the second idler (the lower one) to

3: continue towards the front of the printer

Finally, use the lower idler inside the right Y axis carriage to turn 90° and head to the right side of the printhead carriage

You will need to use whichever toolhead carriage you have chosen to hold both extremities of the belt



As an example, here are both sides of the belt inserted in a Dragonburner MGN9H carriage

Once secured, cut the belt to the correct length, and unmount it to use as reference

Place the freshly cut A belt on top of the remaining GT2 length and make sure they stay perfectly flush (you can tape them together)

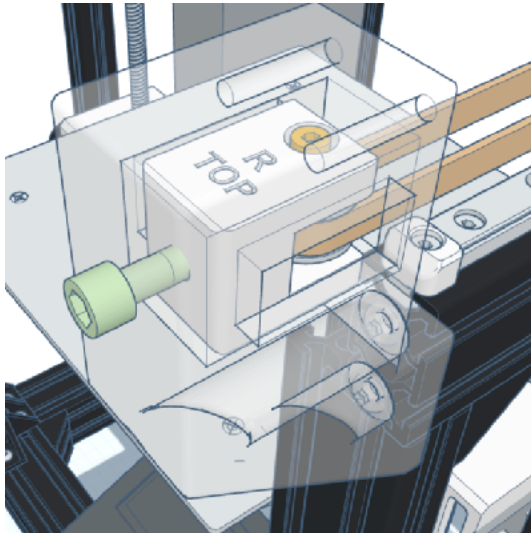
Cut the second belt at the exact dimension, the same teeth count now means the same tension later

You can now reinstall the A belt with the same exact procedure you just followed

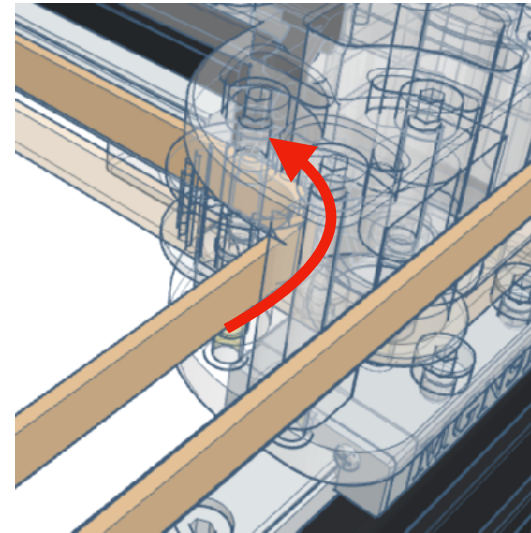
A & B belts installation

B belt

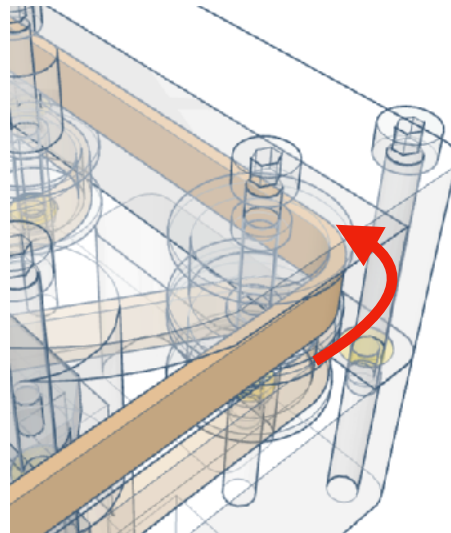
35



Pass the belt into the R side tensioner (flat side against idler) then insert the tensioner into the R front holder, and secure it with a M5x25 SHCS passing from the front hole of the holder



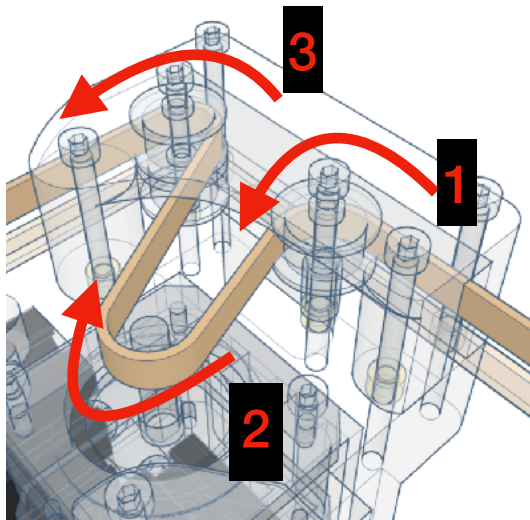
Pass the short side of the belt into the toothed idler and towards the X axis toolhead carriage



The longer side passes next to the right Y carriage all the way to the AB motor holder at the back right
Use the upper idler to turn 90° and head to the left AB motor holder

A & B belts installation

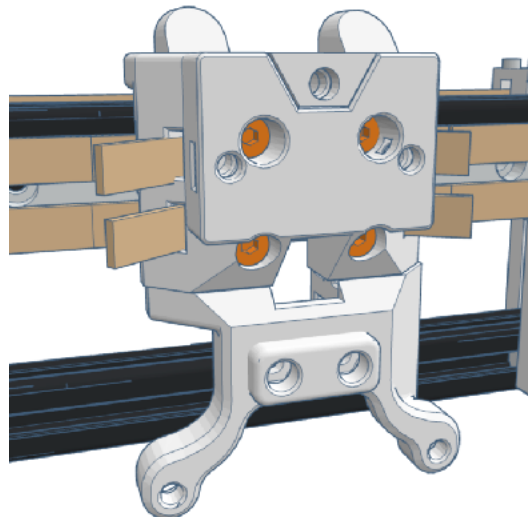
36



1: Into the left AB motor holder, use the first idler to redirect the belt towards the motor pulley (not shown here)

2: Place the belt into the B motor pulley, **it's time to adjust pulley height**. Then back to the second idler (the upper one)

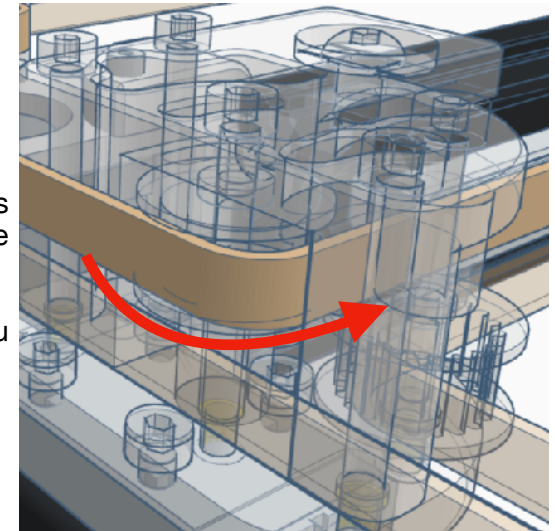
3: continue towards the front of the printer



As an example, here are both belts inserted in a Dragonburner MGN9H carriage, ready to roll

Finally, use the upper idler inside the left Y axis carriage to turn 90° and head to the left side of the printhead carriage

You will need to use whichever toolhead carriage you have chosen to hold both extremities of the belt



IMPORTANT: Double check that both belts are perfectly level and straight all along their path

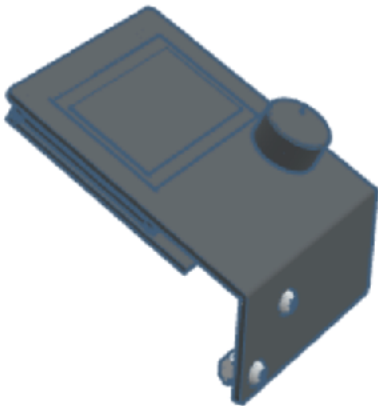
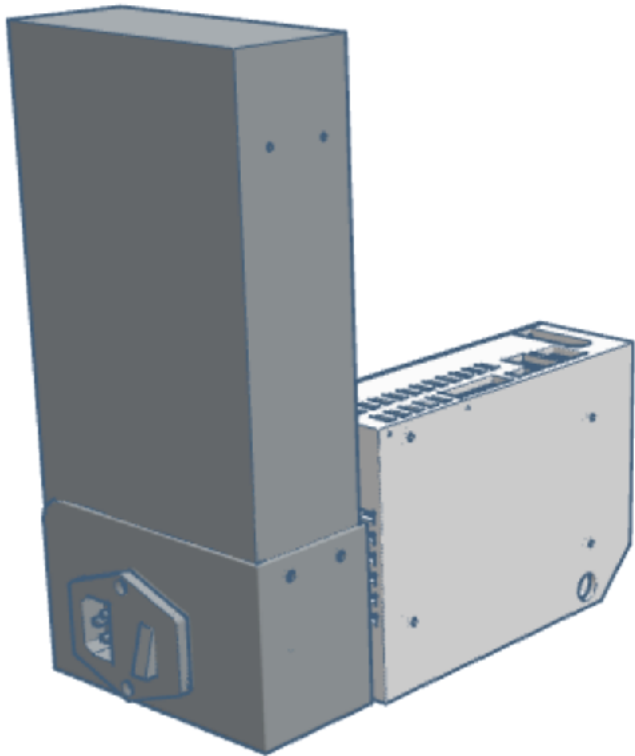
If that is not the case, you may need to adjust the number of M3x0.5 mm spacers used to level the idlers

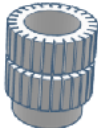
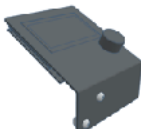
Also check for the A and B motors pulleys height. You may have to unscrew them a bit to align them with the belts

It may be a good idea to add a drop of threadlocker to the grub screws of the pulleys before final tightening

PSU, control box & screen

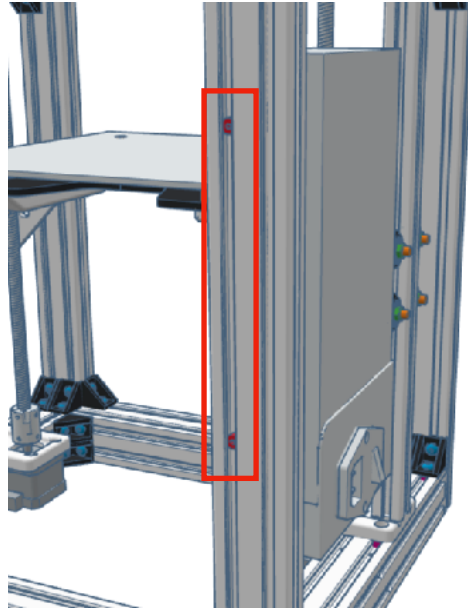
Power supply, control box and screen housings



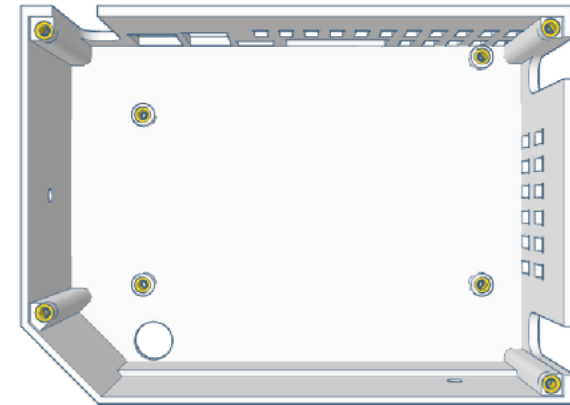
	M3 heat-set insert	8
	M5x8 BHCS	4
	M4x20 SHCS (from Ender 3)	2
	12864 display assembly (from Ender 3)	1
	24V 15A PSU (from Ender 3)	1

PSU, control box & screen

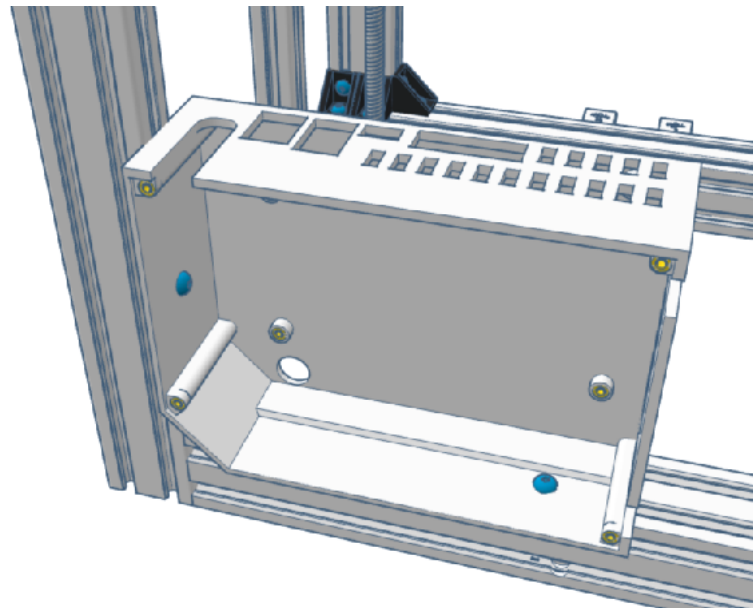
38



Use original M4x20 SHCS to attach the original power supply to the back left 2040-B Z post

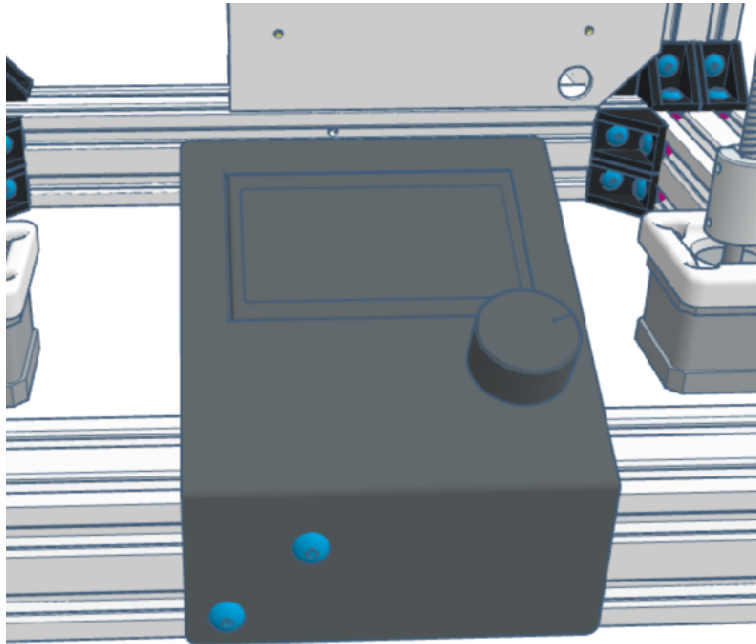


Prepare the bottom of the electronics box by inserting 8 heat-set inserts, one into each corner hole, and on the inside mainboard pillars



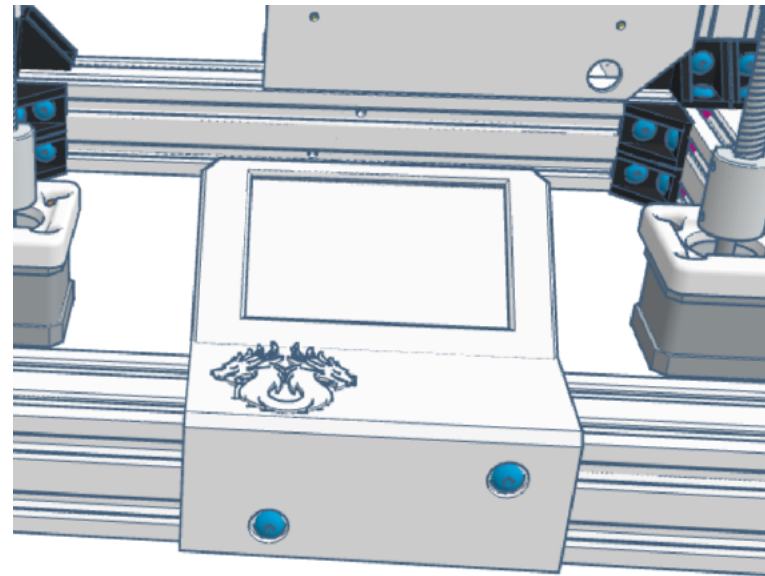
Place the electronics box in the back lower right corner of the frame, and use the two remaining M5 T-slot nuts to secure it with 2 M5x8 BHCS

Box shown is for BTT Manta 5P mainboard. You may need other model depending on your choices



You may use the original Ender 3 12864 screen

It can be mounted to the front of your Duender using the last two remaining M5 T-slot nuts with M5x8 BHCS



The original Duender build however is made with a BTT TFT 35 SPI screen

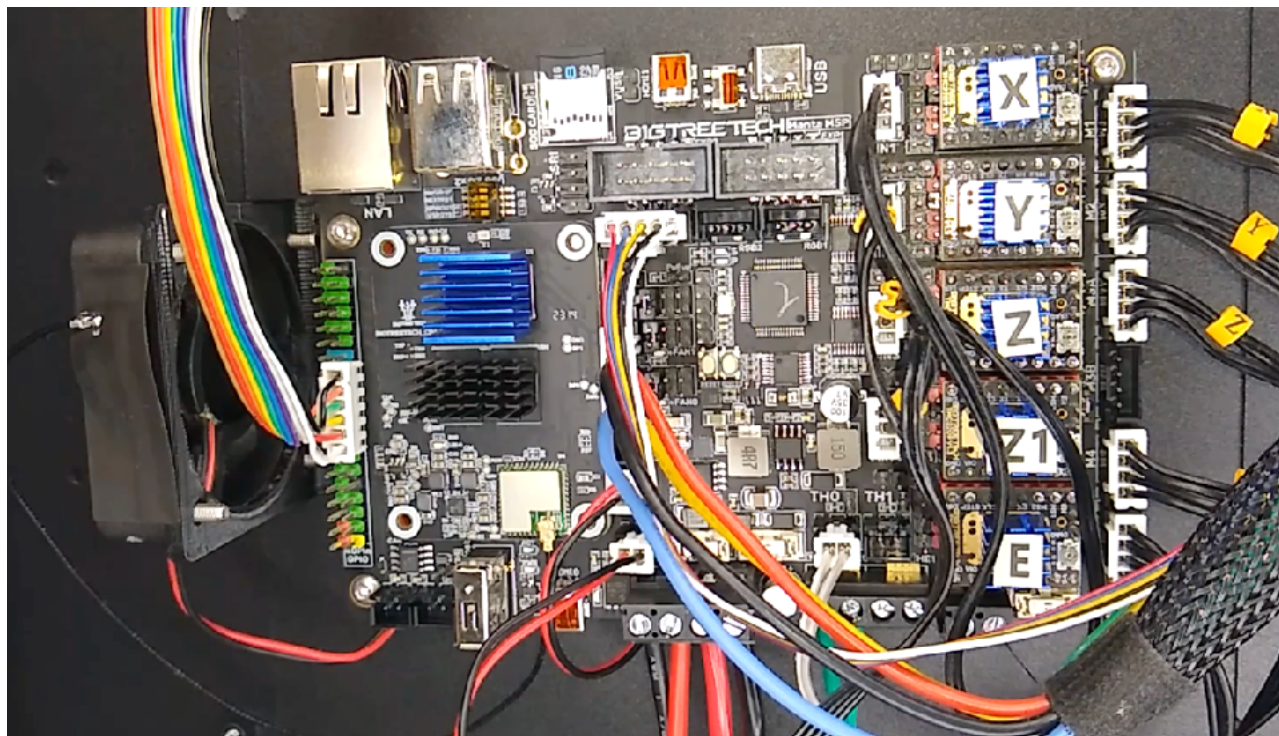
It can be mounted to the front of your Duender using the last two remaining M5 T-slot nuts with M5x8 BHCS

THIS SOLUTION IS NOT RECOMMENDED, as this touch screen seems sluggish. You can replace it with any Klipper compatible screen. Keep in mind that by doing so you will need a specific screen housing

Mainboard installation

Wiring

40



Reference image from Victor Bared's Manta 5P video series.

Jumpers setup, image install: <https://2cm.es/1IYBp>

Wiring: <https://2cm.es/1IYBr>

This section is intended for the BTT Manta 5P mainboard loaded with a CB1/CB2

You will find all the information you need on the BTT Wiki: <https://2cm.es/1IYBC>

Files and firmware can be downloaded from the GitHub: <https://2cm.es/1h2DL>

Keep in mind you may need to extend or completely remake some cables, and for this some tools like a soldering iron, good crimping pliers and a set of connectors will prove more than useful



M3x8 SHCS

8



M3x18 SHCS

8



M3 Nylstop nut

8



BTT Manta M5P
or M8P for Medusa

1



BTT CB1 OR CB2

1



TMC2209 driver

5



4010 axial fan
(from Ender 3)

2



Wires in different colours
1 mm², 0.5 mm² etc.

plenty

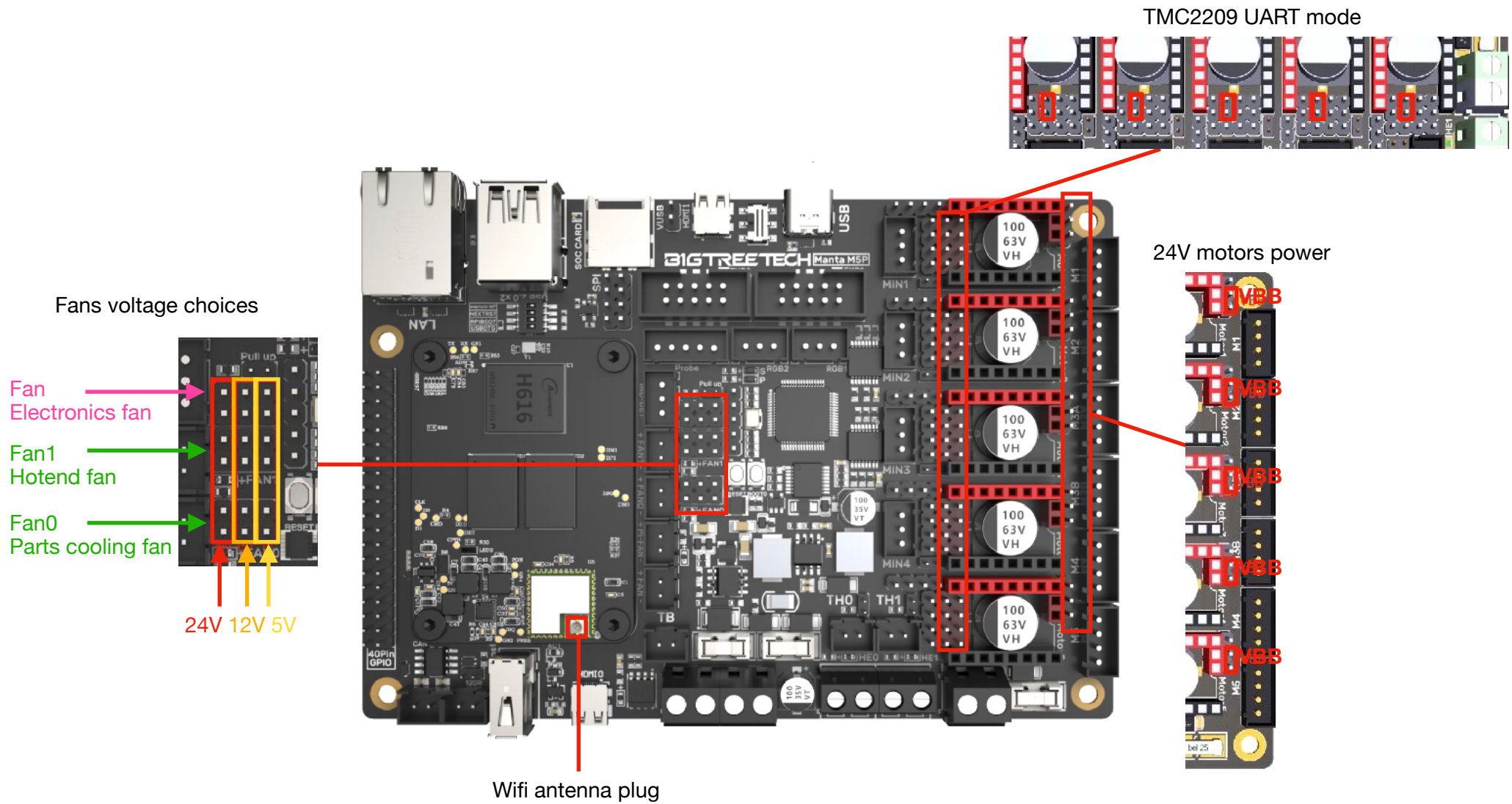
Micro-JST connectors
and pins

some

Mainboard installation

Jumpers basics

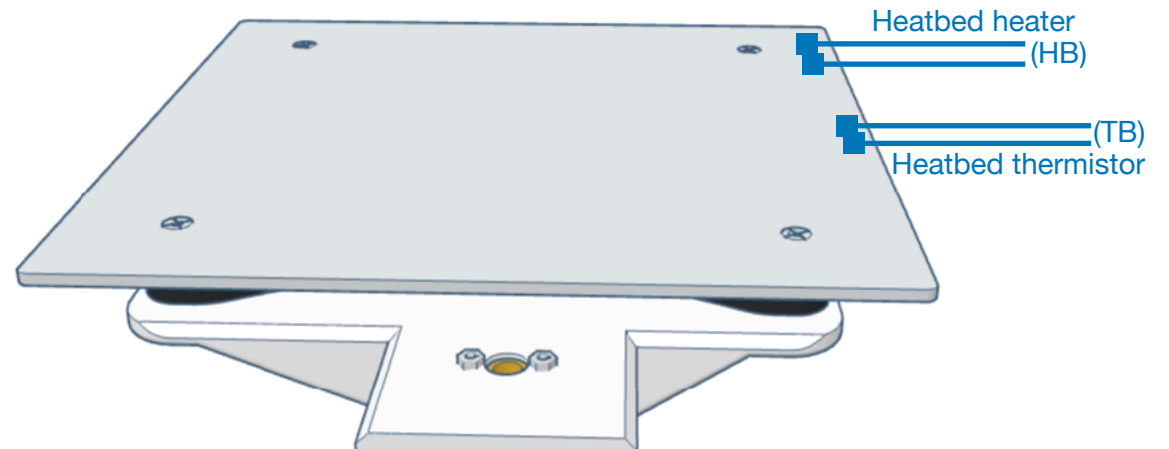
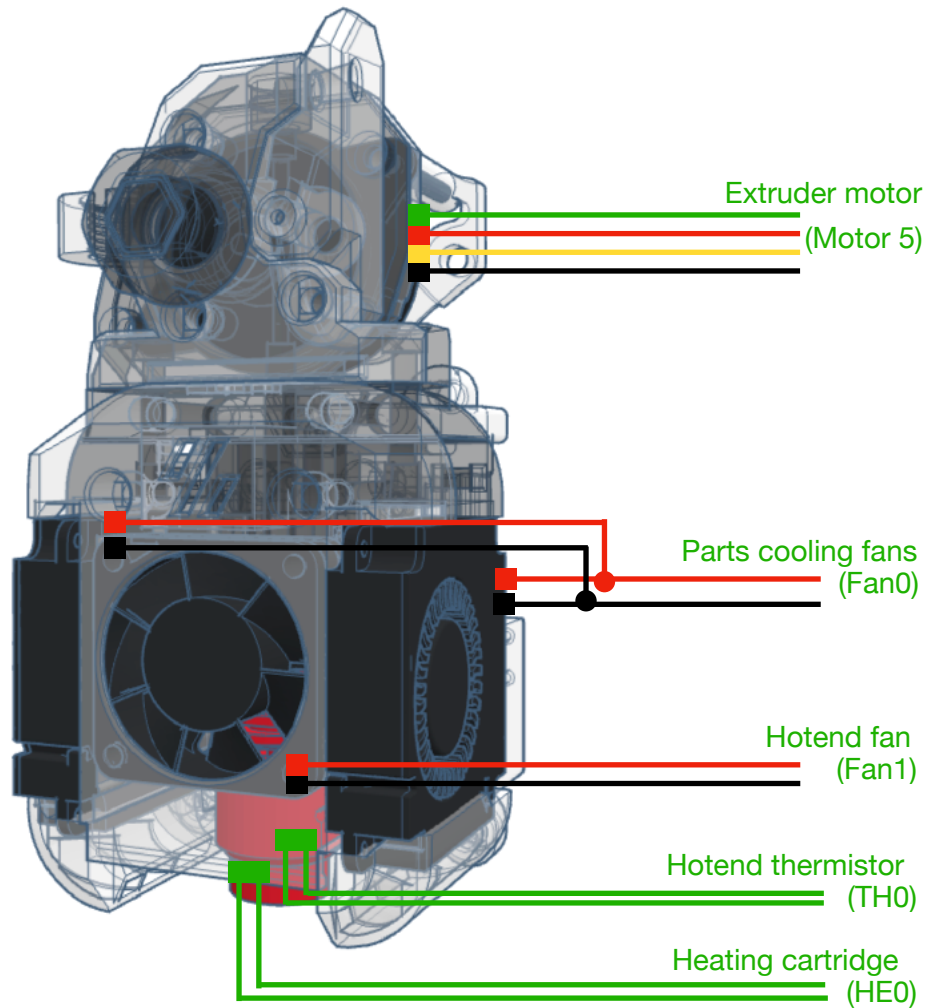
41



Mainboard installation

Toolhead and heatbed wires

42

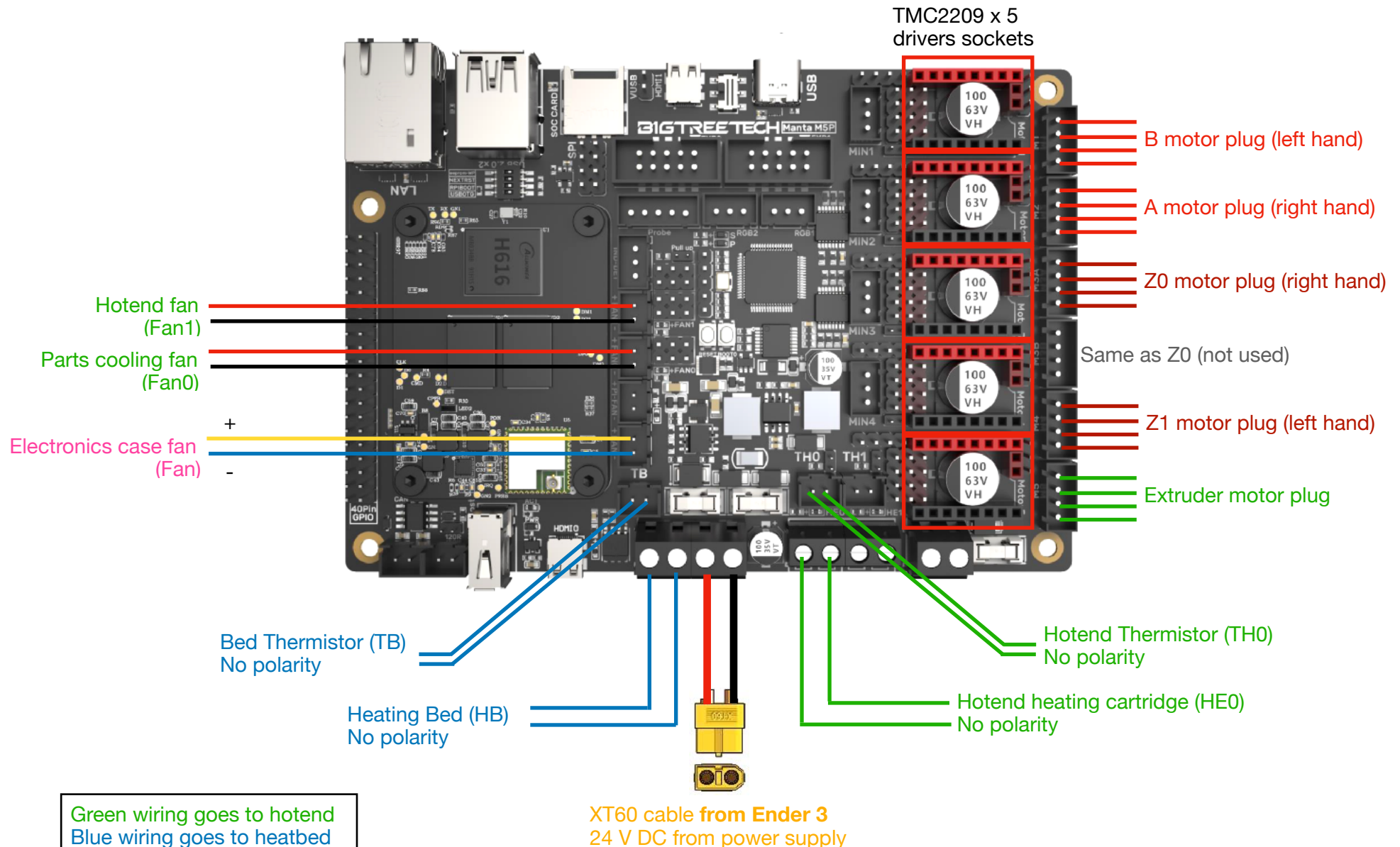


As an example, a Dragonburner toolhead with a Galileo2 extruder and a E3D Voron hotend
However, any toolhead configuration should have the same connectors to the mainboard

Mainboard installation

Mainboard ports

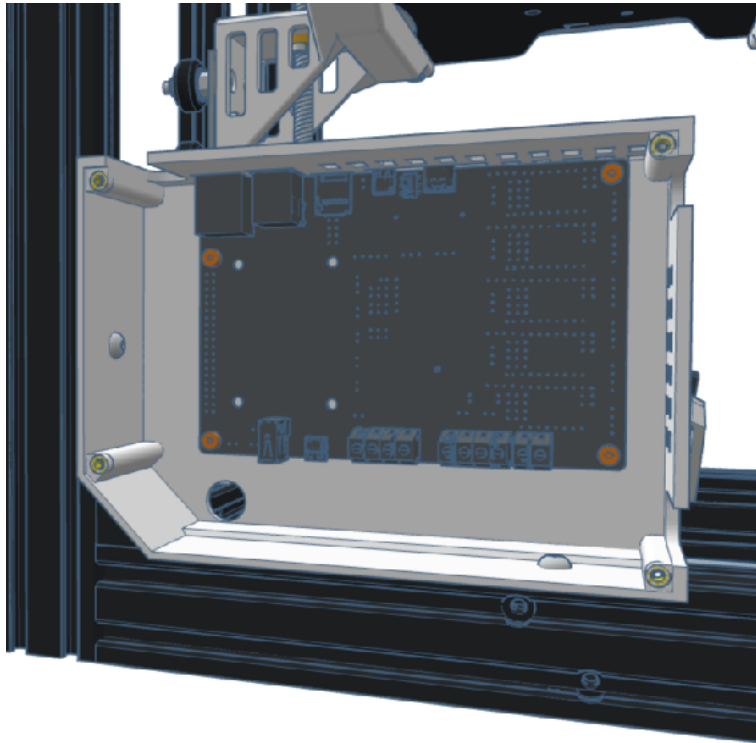
43



Mainboard installation

Going home

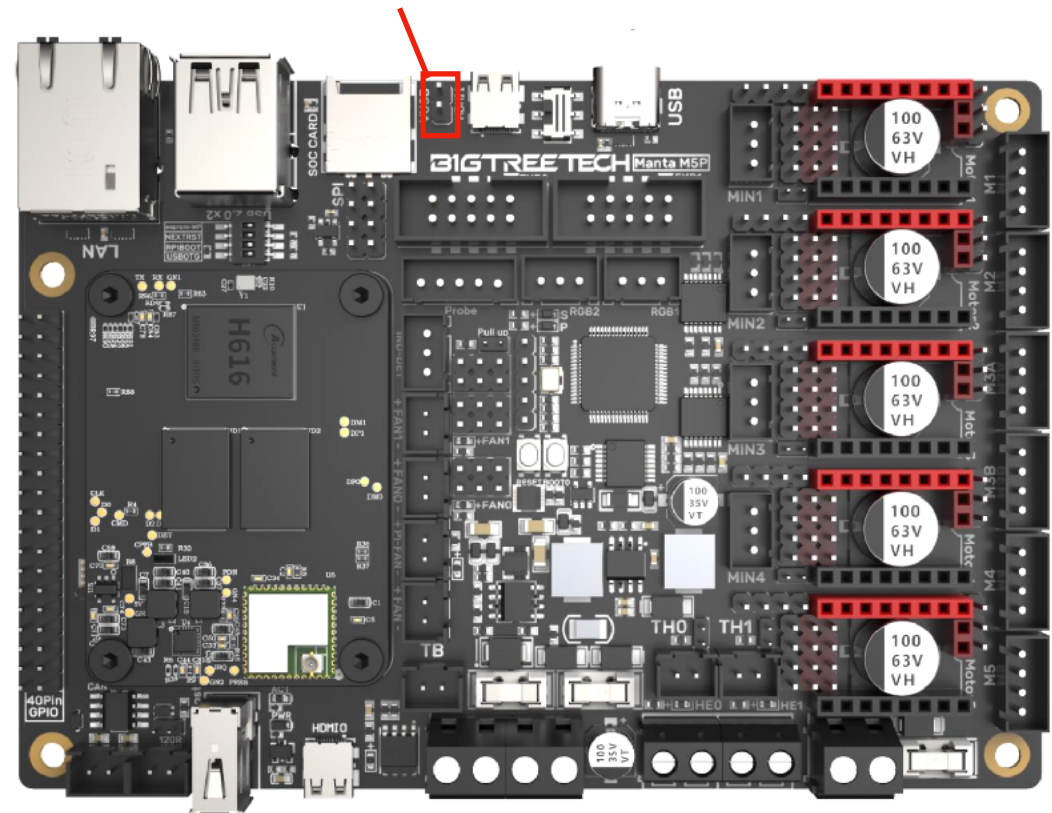
44



You should now be ready to insert your Manta 5P into the electronics control box
Secure it with 4 M3x8 SHCS, and go ahead with all your connections

Make good use of all the box cutouts to manage your cables and secure them with zipties or other sleeving of your choice

Temporary jumper to power the board by USB



To flash the image onto the CB1 microSD card, you may need to use the shown jumper and power the mainboard with 5V by USB cable

Mainboard installation

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(Nearly) Everything you ever wanted to know about Klipper



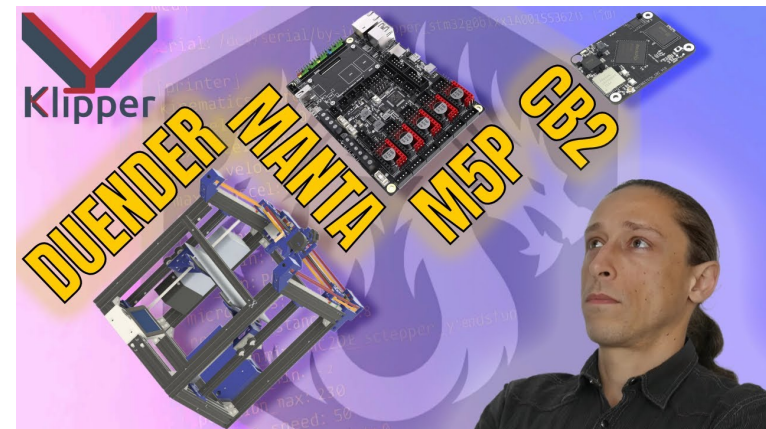
<https://2cm.es/1n0Yt>



<https://2cm.es/1i1Yq>



<https://2cm.es/1i1XM>

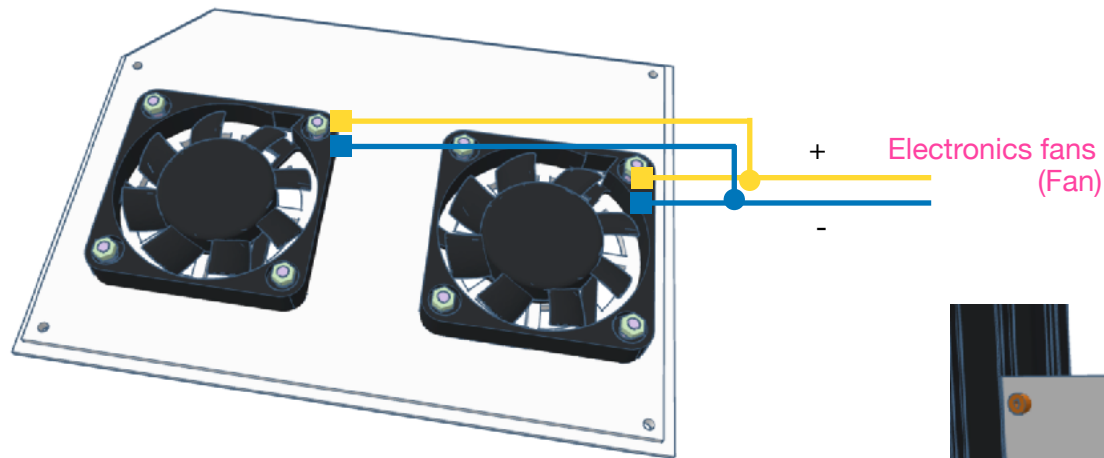


<https://2cm.es/1n0Wo>

Mainboard installation

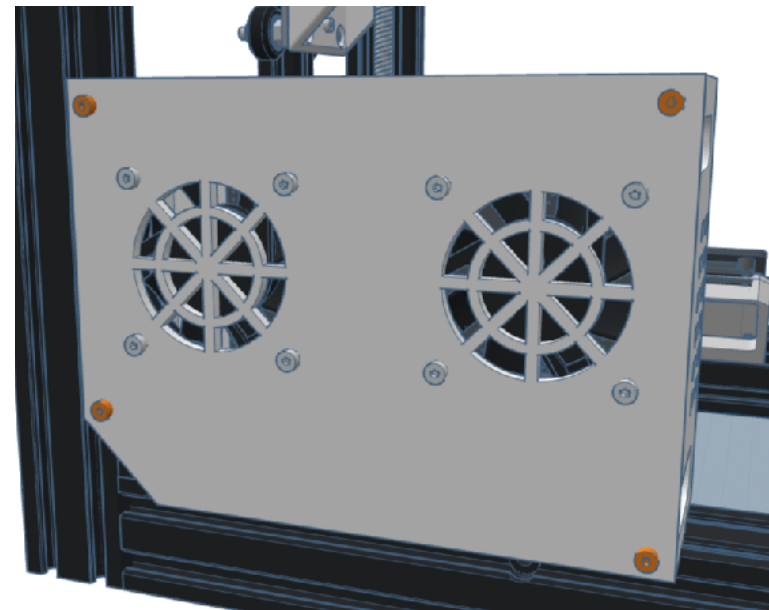
Electronics case fans

46



You may want to cool down your electronics with two 4010 axial fans taken **from the E3 toolheads**

You can use 8 M3x18 SHCS with 8 M3 nuts to secure them to the box lid



Once you are all done, use the last 4 M3x8 SHCS to close the electronics box

Congratulations! Your build is now complete! Next step is installing Klipper and building your printer.cfg file
Once your printer is alive, we can only recommend the excellent [Ellis' Print Tuning Guide](https://ellis3dp.com/Print-Tuning-Guide/): <https://ellis3dp.com/Print-Tuning-Guide/>
We wish you loads of prints and fun with your Duender!

Coming soon!

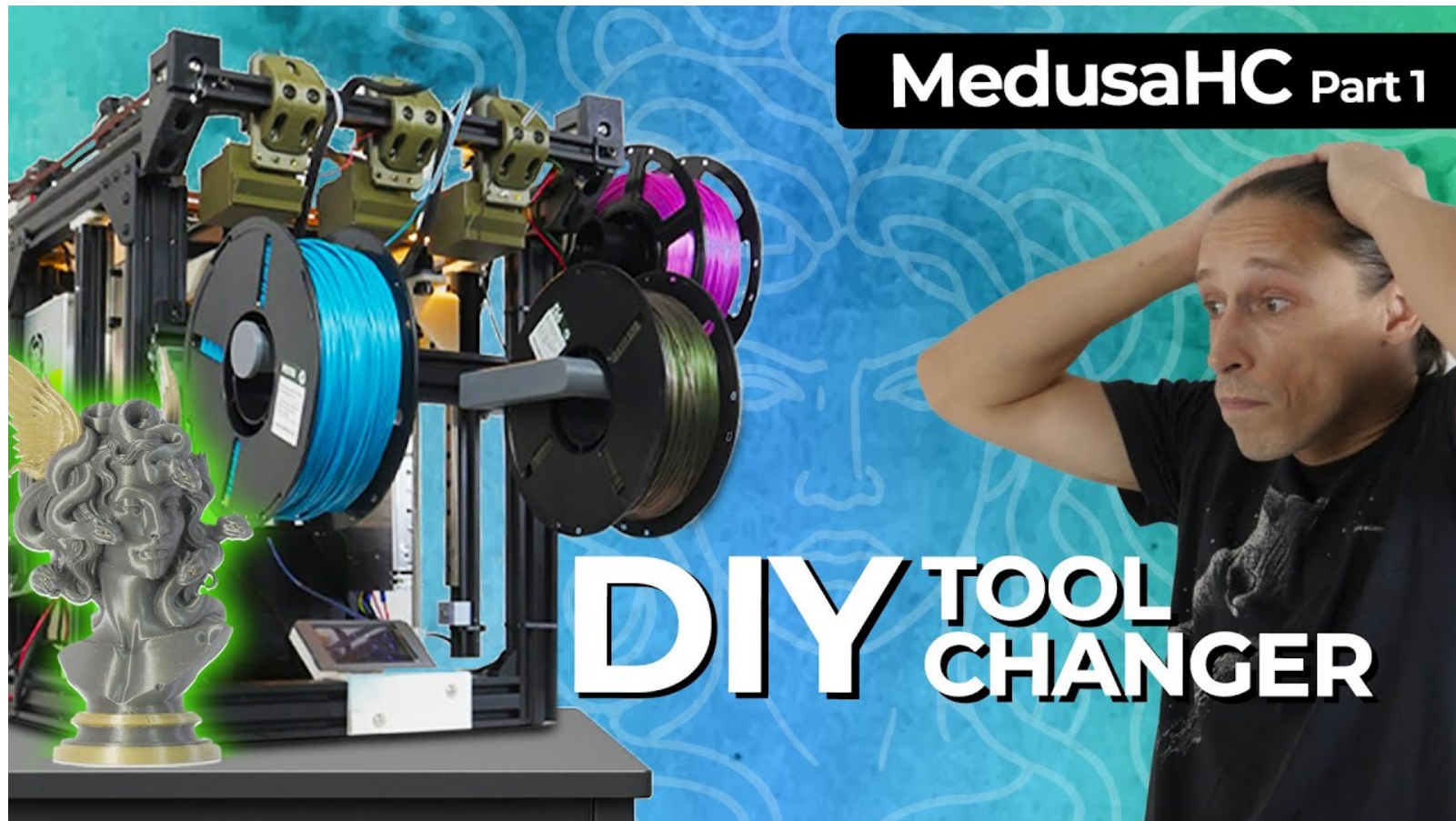
A few words about Medusa HC project

What? You want **more**?

Ok... What if I told you that your Duender can have **up to 4 different toolheads**?
What if you could print 4 different colours or even 4 different materials at the same time, without needing to poop-purge?

Sounds great, doesn't it? Well stay tuned, because your **open-source, budget multi-toolhead mod for the Duender** is coming to you.

More info on Irbis 3D channel : <https://2cm.es/1h7sX>



<https://youtu.be/hpV5Z1TnGdY?si=FyJBHvE2mylvz7cY> or <https://2cm.es/1ml62>